

AIR REGULATIONS



VINOD YADAV AVIATION ACADEMY

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INTERNATIONAL ORGANISATION AND CONVENTION

CHICAGO CONVENTION 1944

Chicago convention consisting of 96 articles was signed on 7 Dec 1944. India has ratified this convention.

Civil and state aircraft

A] This Convention shall be applicable only to civil aircraft and shall not be applicable to state aircraft.

B] Aircraft used in military, customs and police services shall be deemed to be state aircraft

FLIGHT OVER TERRITORY OF CONTRACTING STATES

Right of non-scheduled Flights- Under article 5, aircraft engaged in non-scheduled flights enjoy the right to fly into or across the territory of another State, and to make stops for non-traffic purposes.

Cabotage- Each contracting State shall have the right to refuse permission to the aircraft of other contracting States to take on in its territory passengers, mail and cargo carried for remuneration or hire and destined for another point within its territory.

Applicability of Air Regulations- All aircrafts are to be treated equal while operating from or in the territory of a national, irrespective of their nationality.

Rules of the Air- Over the high seas, the rules in force shall be those established under this Convention.

Prevention of Spread of Disease

Search of Aircraft- The appropriate authorities of each of the contracting State shall have the right without unreasonable delay, to search aircraft of the other contracting States on landing or departure and to inspect the certificates and other documents prescribed by this Convention.

NATIONALITY OF AIRCRAFT

Dual Registration- An aircraft cannot be validly registered in more than one State, but its registration may be changed from one State to another.

Display of Marks- Every aircraft engaged in international air navigation shall bear its appropriate nationality and registration marks.

Report of Registrations- The registering State must report to ICAO data revealing the ownership and control of aircraft it registers.

MEASURES TO FACILITATE AIR NAVIGATION

Aircraft in Distress- Each contracting State undertakes to provide such measures of assistance to aircraft in distress in its territory as it may find practicable.

Investigation of Accidents- In the event of an accident to an aircraft of a contracting State occurring in the territory of another contracting States, and involving death or serious injury, or indicating serious technical defect in the aircraft or air navigation facilities, the State in which the accident occurs will institute as laws permit, with the procedure which may be recommended by the International Civil Aviation Organization. The State in which the aircraft is registered shall be given the opportunity to appoint observers to be present at the inquiry and the State holding the inquiry shall communicate the report and findings in the matter to that State.

CONDITIONS TO BE FULFILLED WITH RESPECT TO AIRCRAFT

Documents Carried in Aircraft-

- A] Its certificate of registration.
- B] Its certificate of airworthiness.
- C] The appropriate licenses for each members of the crew.
- D] Its journey log book.
- E] If it is equipped with radio apparatus, the aircraft radio station licence.
- F] If it carries passengers, a list of their names and places of embarkation and destination.
- G] If it carries cargo, a manifest and detailed declarations of the cargo.

Aircraft Radio Equipment- The aircraft shall carry radios licensed and used in accordance with regulations of the state in which the aircraft is registered.

Licenses of Personnel-

Each contracting State reserves the right to refuse to recognize, for the purpose of flight above its own territory, certificates of competency and licenses granted to any of its nationals by another contracting State.

Journey Log Books- A journey log book in which shall be entered particulars of the aircraft, its crew and of each journey, in such form as may be prescribed from time to time pursuant to this Convention.

Photographic Apparatus- Each contracting state may prohibit or regulate the use of photographic apparatus in aircraft over its territory.

INTERNATIONAL STANDARDS AND RECOMMENDED PRACTICES

Departures from International Standards and Procedures –

Give immediate notification to the International Civil Organisation of the difference between its own practice and that established by the international standard.

Give notice to the council within 60 days of the adoption of the amendment to the international standard.

ICAO

The International Civil Aviation Organisation, a UN Specialized Agency, is the global forum for civil aviation. ICAO works to achieve its vision of safe, secure and sustainable development of civil aviation through cooperation amongst its member States.

Name and composition

Formed by the Chicago Convention 1944. It is made up of an Assembly, a Council, and such other bodies as may be necessary.

ASSEMBLY

All Contracting States one member one vote

COUNCIL

36 Contracting States elected by the Assembly

(President of the Council is elected by the council)

COMMISSIONS AND THE SECRETARIAT COMMITTEES

(Each of between nine and Fifteen members)

(Secretary – General appointed the council)

- | | |
|---|--|
| - Air Navigation commission
19 members appointed by the Council | - Air Navigation
Bureau |
| - Air Transport Committee
Appointed by the Council | - Air Transport
Bureau |
| - Legal Committee
Bureau | - Technical assistance |
| - Committee on joint Support of Air
Navigation Services
Not more than 11 members, with not less
9 members appointed by the council | - Legal Bureau |
| - Finance Committee
Not more the 13 members, with not less
Than 9 members, appointed by the Council | - Bureau of
Administration and
Services. |
| - Committee on Unlawful Interference
15 members appointed by the Council. | |

ICAO has seven regional offices serving nine regions.

Procedures for Air Navigation Services (PANS) – are approved by the Council for worldwide application.

ICAO Annex

Annex 1	Personnel Licensing
Annex 2	Rules of the Air
Annex 3	Meteorological Service for International Air Navigation
Annex 4	Aeronautical Charts
Annex 5	Units of Measurement to be Used in Air and Ground Operations
Annex 6	Operation of Aircraft
Annex 7	Aircraft Nationality and Registration Marks
Annex 8	Airworthiness of Aircraft
Annex 9	Facilitation
Annex 10	Aeronautical Telecommunications
Annex 11	Air Traffic Services
Annex 12	Search and rescue
Annex 13	Aircraft Accident and Incident Investigation
Annex 14	Aerodromes
Annex 15	Aeronautical Information Services
Annex 16	Environment Protection
Annex 17	Security: Safeguarding International Civil Aviation Against Acts of Unlawful Interference
Annex 18	The Safe Transport of dangerous Goods By Air
Annex 19	Safety Management System

ICAO IMPORTANT DOCS.

Doc 4444 – Procedure for air Navigation Services – Air traffic Management (PANS ATM).

Doc.8168 – Aircraft Operations

THE INTERNATIONAL AIR SERVICES TRANSIT AGREEMENT, 1944

The Freedoms of the Air

The Air Services Transit Agreement established to technical freedoms of the air (known as the first and second freedoms)

The Technical Freedom

- a. The First Freedom: The privilege to fly the territory of another participating state without landing.
- b. The Second Freedom: The privilege to land in another participating state for the non-traffic purposes (i.e. refuelling or repair) but not for uplift or discharge of traffic (passengers, cargo or mail).
- c.
- d. **The Commercial Freedoms**
 - a. The Third Freedom: The privilege to put in another state (for example the USA), passengers, mail and cargo taken on in the state of registration (e.g. the UK)
 - b. The Fourth Freedom: The privilege to take on in another state (e.g. the USA passengers mail and cargo destined for the state of airline registration 9eg the UK).
 - c. The Fifth Freedom: The privilege for an airline registered in one state (e.g. the UK) and en-route to or from that state, to take on passengers, mail and cargo in a second state (e.g. Greece) and put them down in third state (e.g. Italy).
 - d.

Modern Freedoms. Due to the process of growth in air transport and the evolution of airlines operating on a global basis, further commercial freedoms have evolved.

- a. The Sixth Freedom: The privilege for an airline registered in one participating state to take on passengers, mail and cargo in a second state, transport them via the state of registration, and put them down in third participating state.
- b. The Seventh Freedom: The privilege for an airline registered in one participating state to take on passengers, mail and cargo in second participating state and put them down in any other participating state without the journey originating, stopping or terminating in the state of registration.
- c.

The Eighth Freedom. With the establishment of the EU and the associated “open skies” policy which reflects the abolition of land frontiers, customs tariffs and immigration restrictions between EU states, a further freedom became necessary to allow the policy to work. This is the privilege of an aircraft registered in one EU State (e.g. Eire) to pick up passengers, mail and cargo in another EU State (e.g. the UK) and carry the same to a destination within that state (e.g. Ryanair).

The Ninth Freedom (Code Sharing): This freedom is a direct result of the IATA conference of Kuala Lumpur and permits interlining or code sharing. This is a scheduled flight being flown by an operator other than the operator to whom the schedule has been granted or with whom the schedule is shared. In this situation, the flight code (identifying the carrier/operator and the schedule flight) is used by another operator. In this situation, the passenger must be informed who the actual carrier is.

UNLAWFUL ACT

CONVENTION ON OFFENCES AND CERTAIN OTHER ACTS COMMITTED ON BOARD AIRCRAFT, SIGNED AT TOKYO, ON 14 SEPTEMBER 1963 (TOKYO CONVENTION)

1. Acts which, whether or not they are offences, may or do jeopardize the safety of the aircraft or of persons or property therein or which jeopardize good order and discipline on board.
2. Convention shall apply in respect of offences committed or acts done by a person on board any aircraft registered in a Contracting State, while that aircraft is in flight or on the surface of the high seas or of any other area outside the territory of any State.

JURISDICTION:

1. The State of registration of the aircraft is competent to exercise jurisdiction over offences and acts committed on board.
2. Each Contracting state shall take such measures as may be necessary to establish its jurisdiction as the state of registration over offences committed on board aircraft registered in such state.

POWERS OF THE AIRCRAFT COMMANDER:

Article 6:

1. The aircraft commander may, when he has reasonable grounds to believe that a person has committed, or is about to commit, on board the aircraft, an offence or act contemplated impose upon such person reasonable measures including restraint.
2. The aircraft commander may require or authorize the assistance of other crew members and may request or authorize, but not require, the assistance of passengers to restrain any person whom he is entitled to restrain.

THE HAGUE CONVENTION OF 1970

The Convention for the Suppression of Unlawful Seizure of Aircraft defines the act of unlawful seizure and the measures to be taken by contracting states to enforce severe punishment upon perpetrators. This agreement specifies extradition of offenders and obliges contracting states to extradite offenders.

THE MONTREAL CONVENTION OF 1971

- Commit acts of violence on board aircraft that endanger people and property and the safety of the aircraft.
- Destroy an aircraft in service or cause damage which renders the aircraft incapable of flight or which is likely to endanger the safety of flight.
- Place a device on board an aircraft that is likely to destroy the aircraft, damage it, or render it unfit for flight.
- Destroy or damage any navigation facility or interference with its correct operation.
- Interfere with aircraft communications or transmit information known to be false that endangers the safety of an aeroplane in flight.

THE MONTREAL CONVENTION 1991

Convention on the marking of plastic explosives for the purpose of detection.

Tokyo Convention 1963, Hague Convention 1970, Montreal Conventions 1971 and 1991 and Montreal Protocol 1988 have been ratified by India.

WARSAW CONVENTION

It unifies legislation on:

- Documentation on the carriage of passengers, baggage and cargo.
- The financial liability of airlines (operations)
- The question of jurisdiction

Liability of the carrier

The Treaty also imposed limitations on the liability of the operator. However, where gross negligence can be proved, the limit of the liability is removed.

The Carriage by Air Act, 1972

The rights and liabilities of air carriers are governed by the Carriage by Air Act, 1972

THE CONVENTION OF ROME 1952

In simple terms, the operator is liable for any damage, but the liability is limited to a sum that is proportionate to the weight of the aircraft. A later Rome Convention looked at the problems of damage caused by foreign aircraft to third parties on the surface of the earth. India has not ratified this convention.

COMMERCIAL PRACTICES AND ASSOCIATED RULES (LEASING)

AOC/AOP - Air Operators Certificate Permit, a document issued by the Authority of a State allowing an Operator to conduct public transport flights.

DRY LEASE – The aeroplane is operated under the AOC of the lessee (the company leasing the aeroplane).

WET LEASE- The aeroplane is operated under the AOC of the Lessor (the company who let the aircraft out).

INDIAN ORGANISATIONS

MINISTRY OF CIVIL AVIATION

Government of India

The ministry of Civil Aviation is responsible for formulation of national policies and programmes for the development and regulation of the Civil Aviation sector in the country. It is responsible for the administration of the Aircraft Act 1934, Aircraft Act 1937.

DGCA is the regulatory body in the field of Civil Aviation primarily dealing with safety issues. It is responsible for regulation of air transport services to/from/within India and for enforcement of civil air regulation, air safety and airworthiness standards. The DGCA also co-ordinates all regulatory functions with the International Civil Aviation Organisation (ICAO)

The Airport Authority of India (AAI) was formed on April 1995 by merging the International Airports Authority of India and the National Airports Authority with a view to accelerate the integrated development, expansion, and modernization of the operational, terminal and cargo facilities at the airports in the country conforming to international standards.

AIRCRAFT NATIONALITY AND REGISTRATION MARKS

Introduction: Annex 7 sets out procedures for selection by ICAO Contracting States of nationality marks from the nationality symbols included in the radio call signs allocated to the States of Registry by the ITU.

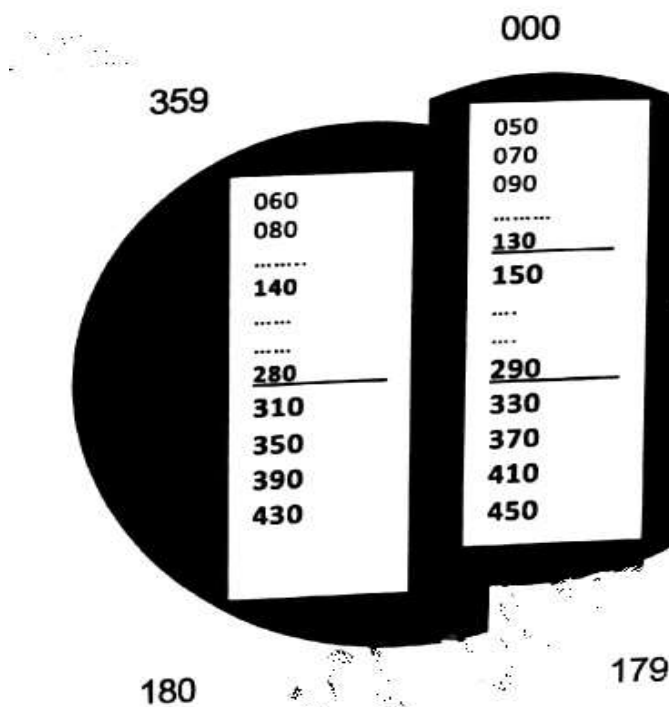
- The registration mark shall be letters, numbers, or a combination of letters and numbers consisting of 1 to 5 digits or characters and shall be that assigned by the State of Registry or common mark registering authority.
- When letters are used for the registration mark, combinations shall not be used which might be confused with the five-letter combinations used in the International Code of Signals, part II, the three letter combinations beginning with Q used in the Q Code, and with the distress signal SOS, or other similar urgent signals, for example XXX, PAN and TTT.

CRUISING LEVEL SYSTEMS

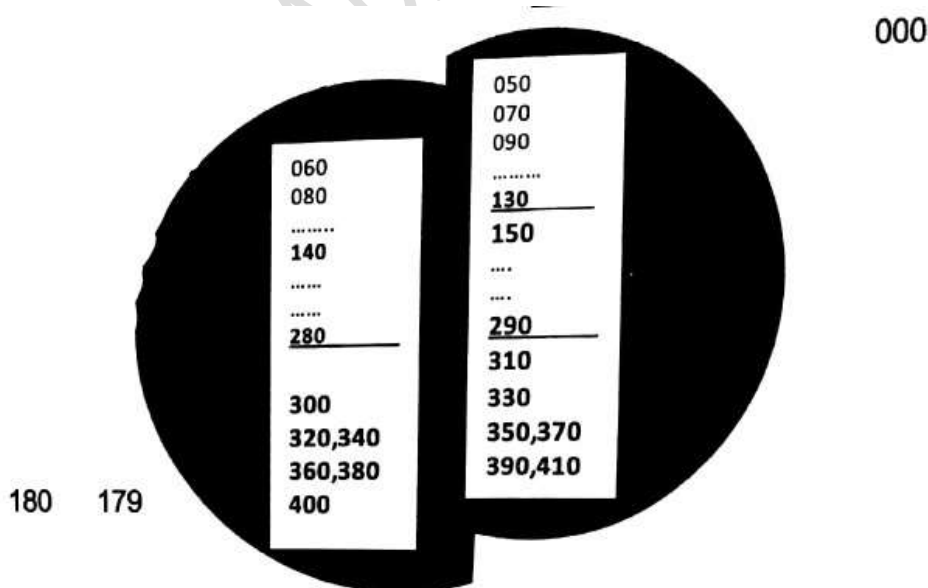
- Based on MAGNETIC TRACKS
- HIGHEST AVAILABLE LEVEL FOR FLYING: FL 450
- MAXIMUM FLIGHT LEVEL: FL 460
- LOWEST FLIGHT LEVEL: FL 050

CONVENTIONAL/STANDARD VERTICAL SEPARATION MINIMA (CVSM)

Semi Circular System



REDUCED VERTICAL SEPARATION MINIMA (RVSM)



NATIONALITY AND REGISTRATION MARKS

Wings- The height of the marks on the wings of heavier-than-air aircraft shall be at least 50 centimetres.

Fuselage (or equivalent structure) and vertical tail surfaces- The height of the marks on the fuselage (or equivalent structure) and on the vertical tail surface of heavier-than-air aircraft shall be at least 30 centimetres.

Lighter-than-air aircraft- The height of the marks on lighter-than-air aircraft other than a unmanned free balloons shall be at least 50 centimetres. The measurement of the marks related to unmanned free balloons shall be determined by the State of Registry.

NATIONAL PROVISIONS

Certificate of Registration- The authority empowered to register aircraft and to grant certificate of registration in India shall be the Central Government.

In the case of a leased aircraft, the certificate of registration shall also include the validity of the lease and the names, nationalities and addresses of the lessor and the lessee.

Register of aircraft: A register of aircraft registered in India shall be maintained by the Director-General.

Nationality and Registration Marks, how to be affixed-

- 1) India has been allotted nationality symbols AT to AW by ITU. However, we continue to use pre- Independence nationality mark capital letters VT in Roman character assigned by the Director-General. Nationality mark of the defence aircraft is capital letters VU in Roman character. A hyphen shall be placed between the nationality mark and the registration mark.

Use of State Marks-

- 1) An aircraft shall not bear on any part of its exterior surface, any advertisement as required or permitted by the Director General.

RULES OF THE AIR

TERRITORIAL APPLICATION OF THE RULES OF THE AIR

These rules are applicable to all aircraft flying over Indian territory and aircraft bearing the Indian nationality and registration marks.

Responsibility for compliance with the rules of the air

Responsibility of pilot-in-command

The pilot-in-command of an aircraft shall have final authority as to the disposition of the aircraft while in command.

Acrobatic Flight

The acrobatic shall be commenced at such a height that will permit completion of the manoeuvre at height of not less than 600 metres (2000 feet) above the ground.

No Person shall Fly Acrobatically:

In the vicinity of an aerodrome at a distance of less than two nautical miles from the nearest point of the perimeter of the aerodrome unless flown at height greater than 1800 meters (6000ft).

Formation Flights

No civil aircraft shall be flown in formation in India.

Right-of-way

The aircraft that has the right-of-way shall maintain its heading and speed.

Approaching Head-on

When two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

Converging:

When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way, except as follows:-

- a) Power-driven heavier-than-air aircraft shall give way to airships, gliders and balloons.
- b) Airships shall give way to balloons.
- c) Gliders shall give way to balloons.
- d) Power-driven aircraft shall give way to aircraft which are seen to be towing other aircraft or objects.

Overtaking: -

An overtaking aircraft is an aircraft that approaches another from the rear on a line forming an angle of less than 70 degrees with the plane of symmetry of the latter. Unable to see either of the aircrafts left (port) or right (starboard) navigation lights.

SURFACE MOVEMENT OF AIRCRAFT

- a) When two aircraft are approaching head on, or approximately so, each shall stop or where practicable alter its course to the right so as to keep well clear.
- b) When two aircraft are on a converging course, the one which has the other on its right shall give way
- c) An aircraft which is being overtaken by another aircraft shall have the right-of-way.

LIGHTS TO BE DISPLAYED BY AIRCRAFT

- a) Anti-collision lights.
 - b) Navigation lights.
-
- 1. Navigation lights. Navigation lights must be displayed from sunset to sunrise or otherwise as specified as per instructions in following manner: -
 - (a) An unobstructed red light projected above and below the horizontal plane through an angle from dead ahead to 110° to the left (port).
 - (b) An unobstructed green light projected above and below the horizontal plane through an angle from dead ahead to 110° to the right (starboard).
 - (c) An unobstructed white light projected above and below the horizontal plane rearward through an angle of 140° equally distributed on the left and right sides.
 - 2. The lights may appear either as steady or as flashing lights and if the flashing system is used, either one or both of the following additional lights may be displayed: -
 - (a) A flashing red rear light which alternates with flashing white rear light.
 - (b) A flashing white light visible in all directions which alternates with the single emitted by the lights described in 1 (a), (b) and (c).
 - 3. Anti-collision light. When the navigation lights are displayed as steady lights, a flashing red light visible in all directions, within 30° above and below the horizontal plane may also be displayed.

SIMULATED INSTRUMENT FLIGHTS

An aircraft shall not be flown under simulated instrument flight conditions unless: -

- a) Fully functional dual controls are installed in the aircraft.

OPERATION ON AND IN THE VICINITY OF AN AERODROME

Make all turns to the left, when approaching for a landing and after taking off, unless otherwise instructed.

RIGHT OF WAY AT NIGHT

If a pilot sees navigation lights which is not moving relative to him, there is a risk of collision. If the light is green, he has right of way. If the light is red, he must give way by altering course to the right. If he sees both red and green the other aircraft is approaching head on and he must alter course to the right.

UNLAWFUL INTERFERENCE

- a) Proceed in accordance with applicable special procedures for in-flight contingencies, where such procedures have been established and promulgated in the Regional Supplementary Procedures.
- b) If no applicable regional procedures have been established, proceed at a level which differs from the cruising levels normally used for IFR flight by :
 - 1) 150 m (500ft) in an area where a vertical separation minimum of 300 m (1000ft) is applied.
 - 2) 300 m (1000ft) in an area where a vertical separation minimum of 600 m (2000ft) is applied.

FLIGHT PLANS

1. ICAO Flight Plan form

International Flight Plan			
PRIORITY Priorité <<< FF >>>		ADDRESSEE(S) Destinataire(s) _____ _____	
FILING TIME Heure de dépôt _____		ORIGINATOR Expéditeur _____	
SPECIFIC IDENTIFICATION OF ADDRESSEE(S) AND/OR ORIGINATOR Identification précise du(des) destinataire(s) et/ou de l'expéditeur _____			
3 MESSAGE TYPE Type de message <<< (FPL) >>>		7 AIRCRAFT IDENTIFICATION Identification de l'aéronef _____	
9 NUMBER Nombre _____		8 FLIGHT RULES Règles de vol _____	
13 DEPARTURE AERODROME Aéroport de départ _____		10 EQUIPMENT Équipement _____	
15 CRUISING SPEED Vitesse croisière _____		TIME Heure _____	
LEVEL Niveau _____		ROUTE Route _____	
16 DESTINATION AERODROME Aéroport de destination _____			
16 OTHER INFORMATION Renseignements divers _____		TOTAL EET Durée totale estimée HR. MIN _____	
ALTN AERODROME Aéroport de dégagement _____		2ND. ALTN AERODROME 2 ^e aéroport de dégagement _____	
SUPPLEMENTARY INFORMATION (NOT TO BE TRANSMITTED IN FPL MESSAGES) Renseignements complémentaires (À NE PAS TRANSMETTRE DANS LES MESSAGES DE PLAN DE VOL DÉPOSÉ)			
19 ENDURANCE Autonomie HR. MIN _____		PERSONS ON BOARD Personnes à bord _____	
SURVIVAL EQUIPMENT Équipement de survie _____		JACKETS/Gilets de sauvetage _____	
POLAR Pôle _____		DESSERT Désert _____	
MARITIME Maritime _____		JUNGLE Jungle _____	
DINGHIES/Canots _____		LIGHT Lampes _____	
NUMBER Nombre _____		FLUORESC Fluores _____	
CAPACITY Capacité _____		UHF UHF _____	
COVER Couverture _____		VHF VHF _____	
COLOUR Couleur _____		ELT ELT _____	
AIRCRAFT COLOUR AND MARKINGS Couleur et marques de l'aéronef _____			
REMARKS Remarques _____			
PILOT-IN-COMMAND Pilote commandant de bord _____			
FILED BY/Déposé par _____			
SPACE RESERVED FOR ADDITIONAL REQUIREMENTS _____			

2.1 General

Adhere closely to the prescribed formats and manner of specifying data.

Commence inserting data in the first space provided. Where excess space is available leave unused space blank.

Insert all clock time in four figures UTC.

Insert all estimated elapsed time in 4 figures (hours and minutes).

Shaded area preceding item 3 - to be completed by ATS and com services, unless the responsibility for originating flight plan messages has been delegated

Note - the term “aerodrome” where you in the flight plan is intended to cover also sites other than aerodromes which may be used by certain types of aircraft, e.g. helicopters or balloons.

2.2 Instruction for insertion of ATS data

Complete items 7 to 18 as indicated hereunder.

Complete also item 19 as indicated hereunder, when so required by the appropriate ATS authority Or when otherwise deemed necessary.

Note - Item number on the form or not consecutive, as they correspond to field type number in ATS messages.

Item 7 : Aircraft identification (maximum 7 characters)
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Insert one of the following aircraft identifications, not exceeding 7 characters:

The registration marking of the aircraft (e.g. EIAKO, 4XBCD, N2567GA), when:

1. in radiotelephony the call sign to be used by the aircraft will consist of this identification alone (e.g. SABENA OOTEK);
2. the aircraft is not equipped with the radio;

Or B. The ICAI designator for the aircraft operating agency followed by the

flight identification number (e.g. KLM 511, NGA213, JTR25) when in radio telephone the call sign to be used by the aircraft will consist of the ICL telephony designated for the operating agency followed by the flight identification (e.g. KLM 511, NIGERIA123, HERBIE25).

Note- provision for the use of radiotelephony call signs are contained in Annex10, Volume II, chapter 5, ICAO designators, and telephone designators for aircraft operating agencies are contained in Doc 8585 – designators for aircraft operating agencies, Aeronautical authorities and services.

Item 8: FLIGHT RULES AND TYPE OF FLIGHT (1 OR 2 CHARACTERS)

Flight Rules

INSERT one of the following letter to denote the category of flight with which the

pilot intends to comply.

I if IFR

V if VFR

Y if IFR first*

Z if VFR first*

* if including either Y or Z, specify in item 15 the point of points

where a change of flight rules is planned.

Type of Flight

INSERT one of the following letter to denote the type of flight when so required by

the appropriate ATS authority

S if scheduled air transport

N if no scheduled air transport operation.

G in general aviation

M if military

X if other than many of the defined categories above.

ITEM 9 : NUMBER AND TYPE OF AIRCRAFT AND WAKE TURBULENCE DATA

Number of aircraft (1 or 2 characters)

INSERT the number of aircraft, if more than one.

type of aircraft (2 and 4 characters)

INSERT the appropriate designated as specified in ICAO Doc 8643, Aircraft Type Designators,

OR if no such does has been assigned, or in case of formation flights comprising more than one type,

INSERT ZZZZ, and SPECIFY the number(s) and type(s) of aircraft preceded by TYP/.

Wake Turbulence category (1 character)

INSERT and followed by one of the following letters to indicate the wake turbulence category of the aircraft:

H HEAVY, to indicate an aircraft type with a maximum certificate take-off mass of 136,000 kg or more;

M MEDIUM, to indicate an aircraft type maximum certificated take-off mass of less than 136,000 kg more than 7000 kg;

L LIGHT, to indicate an aircraft type with a maximum certificated take-off mass of 7000 kg or less.

ITEM 10: EQUIPMENT

Radio Communication, navigation and approach aid equipment

INSERT one letter as follows:

N if no COM/NAV/ approach aid equipment for the route to be flown is carried, or the equipment is unserviceable,

OR,

S if standard COM/NAV/ approach aid equipment for the route to be flown is carried (See Note 1),

AND/OR

INSERT one or more of the following letters to indicate the COM/NAV/ approach aid equipment available and serviceable:

A (Not allocated) I Inertial Navigation

R RNP type certification

(See Note 5)

B (Not allocated) J (Data Link)

T TACAN

(See Note 3)

C LORAN C

K MLS

U UHF RTF

D DME

L ILS

V VHF RTF

E (Not allocated) M Omega

W when prescribed by ATS

F ADF

O VOR

X when prescribed by ATS

G (GNSS)

P (Not Allocated) Y when prescribed by ATS

H HF RTF

Q (Not Allocated)

Z Other equipment carried

(See Note 2)

Note1 Standard equipment is considered to be VHF RTF, ADF, VOR and ILS, unless another combination is prescribed by the appropriate ATS authority

- Note2 If the letter Z is used, specify in Item 18 the other equipment carried, preceded by COM/and or NAV/ as appropriate.
- Note3 the letter J is used, specify in Item 18 the other equipment carried, preceded by DAT followed by one or more letters, as appropriate.
- Note 4 Information on navigation adaptability provided to ATC for clearance and routing purposes.
- Note5 Inclusion of letter R indicated that an aircraft meets the RNP type prescribed for the route segment(s) and/or route concerned.

Surveillance equipment

INSERT one outward of the following letters to describe the services surveillance equipment carried:

SSR equipment

- N Nil
- A Transponder- Mode A (4 digits – 4,096 codes)
- C Transponder- Mode A (4 digits – 4,096 codes) and Mode C
- X Transponder- Mode S, including pressure-altitude transmission, but no aircraft identification transmission
- I Transponder- Mode S, including aircraft identification transmission, but no pressure – altitude transmission
- S Transponder – Mode S, including both pressure-altitude and aircraft identification transmission

ADS equipment:

- D ADS capability

ITEM:13 DEPARTURE AERODROME AND TIME (8 CHARACTERS)

- INSERT the ICAO four-letter Location Indicator of the departure aerodrome,
- OR if no Location Indicator has been assigned,
- INSERT ZZZZ, and SPECIFY, in Item 18, the name of the aerodrome, preceded by DEP/,
- OR if the flight plan is received from an aircraft in flight.
- INSERT AFIL, and SPECIFY, in Item 18, the ICAO four letter Location Indicator of the location of the ATS unit from which supplementary flight plan data can be obtained, preceded by DEP/,
- THEN, WITHOUT A SPACE
- INSERT for a flight plan submitted before departure on the same day, use estimated 4-digit off-block time (HHMM),
- OR, for a flight plan that will be activated on a different day, use a 6-digit date-time group; the first 2 digits will be the date and the next 4 digits will be the proposed departure time (DDHHMM),

ITEM 15: ROUTE

INSERT the first cruising speed as in (a) and the first cruising level as in (b), without a space between them.

THEN, following the arrow, INSERT the route description as in (c).

(a)Cruising speed (maximum 5 characters)

INSERT the True Air Speed for the first or the whole cruising portion of the flight, in terms of:
Kilometres per hour, expressed as K followed by 4 figures (e.g. K0830), or

Knots, expressed as N followed by 4 figures (e.g. N0485), or

Mach Number, when so prescribed by the appropriate ATS, to the nearest hundredth of unit Mach, expressed as M followed by 3 figures (e.g. M082).

(b) Cruising level (maximum 5 characters)

INSERT to planned cruising level for the first or the whole portion of the route to be flown, in terms of:
Flight Level, expressed as F followed by 3 figures (e.g. F085; F330); or

*Standard Metric Level in tens of meters, expressed as S followed by 4 figures (e.g.S1130); or

Altitude in hundreds of feet, expressed as A followed by 3 figures (e.g. A045; A100); or

Altitude in tens of meters, expressed as M followed by 4 figures (e.g. M0840), or for uncontrolled VFR flights the letters VFR.

*When so prescribed by the ATS authorities.

(c)Route (including changes of speed, level and/or flight rules)

Flights along designated ATS routes

INSERT if the departure aerodrome is located or connected to the ATS route, the designator of the first ATS route,

OR if the departure aerodrome is not on or connected to the ATS route, the letters DCT followed by the point of joining the first ATS route, followed by the designator of the ATS route.

THEN,

INSERT each point at which either a change of speed or level, a change of the ATS route, and/or change of flight rules is planned,

Note – When a transition is planned between a lower and upper ATS route and the routes are oriented in the same direction, the point of transition need not be inserted.

FOLLOWED IN EACH CASE by the designator of the next ATS route segment, even if the same as the previous one,

OR by DCIT, if the flight to the next point will be outside a designated route, unless both Defined by geographical coordinates.

Flights outside designated ATS routes

INSERT points normally not more than 30 minutes flying time or 370 km (200 NM) apart, Including each point at which a change of speed or level, a change of track, or a change of flight rules is planned.

OR when required ATS authority (ies),

DEFINE the track of flights operating predominantly in an east-west direction between 70°N and 70°S by reference to significant points formed by the intersections of half or whole degree of latitude with meridians spaced at intervals of 10° of longitude. For flights operating in areas outside those latitudes the tracks shall be defined by significant points formed by the intersection of parallels of latitude with meridians normally spaced at 20° of longitude. The distance between significant points shall, as far as possible, not exceed one hour's flight time. Additional significant points shall be established as deemed necessary.

For flights operating predominantly in a north-south direction, define tracks by reference to significant points formed by the intersection of whole degrees of longitude with specified parallels of latitude which are spaced at 5°

INSERT DCT between successive points unless both points are defined by geographical coordinates or by bearing and distance.

USE ONLY the conventions in (1) to (5) below and SEPARATE each sub-item by a space.

(1) ATS route (2 to 7 characters)

The coded designator assigned to the route or route segment including, where appropriate, the coded designator assigned to the standard departure or arrival route (e.g. 3CKI, B1, R14, UB10, KODAP2A).

(2) Significant point (2 to 11 characters)

The coded designator (2 to 5 characters) assigned to the point (e.g. LN, MAY, HADDY), or, if no coded designator has been assigned, one of the following ways:

Degrees only (7 characters):

2 figures describing latitude in degrees, followed by "N" (North) or "S" (South), followed by 3 figures describing longitude in degrees, followed by "E" (East) or "W" (West). Make up the correct number of figures, where necessary, by insertion of zeroes, e.g. 45N078W

Degrees and minutes (11 characters):

4 figures describing latitude in degrees and tens and units of minutes followed by "N" (North) or "S" (South), followed by 5 figures describing longitude in degrees and tens and units of minutes, followed by "E" (East) or "W" (West). Make up the correct number of figures, where necessary, by insertion of zeroes, e.g. 4620N07805W.

Bearing and distance from a navigation aid:

The identification of the navigation aid (normally a VOR), in the form of 2 or 8 characters, THEN the bearing from the aid in the form of 3 figures giving degrees magnetic THEN the distance from the aid in the form of 3 figures expressing nautical miles. Make up the correct number of figures, where necessary, by insertion of zeroes, e.g. a point 180° magnetic at a distance of 40 nautical miles from VOR "DUB" should be expressed DUB 180040.

(3) Change of speed or level (maximum 21 characters)

The point at which a change of speed (5% TAS or 0.01 Mach or more) or a change of level is planned, Expressed exactly as in (2) above, followed by an oblique stroke and both the cruising speed and the cruising level, expressed exactly as in (a) and (b) above without a space between them, even when only one of these quantities will be changed.

Examples:

LN/N0284A045

MAY/N0305F180
HADDY/N0420F330
4602N07805W/N0500F350
46N078W/M082F330
DUB180040/N0350M0840

(4) Change of flight rules (maximum 3 characters)

The point at which the change of flight rules is planned, expressed exactly as in (2) or (3) above as appropriate, followed by a space and one of the following:

VFR if from IFR to VFR
IFR if from VFR to IFR

Examples:

LN VFR
LN/N0284A050 IFR

(5) Cruise climb (maximum 28 characters)

The letter C followed by an oblique stroke: THEN the point at which cruise climb is planned to start, Expressed exactly as in (2) above, followed by an oblique stroke; THEN the speed to be maintained during cruise climb, expressed exactly as in (a) above, followed by the two levels defining the layer to be occupied during cruise climb, each level expressed exactly as in (b) above or the level above which cruise is planned followed by the letters PLUS, without a space between them.

Examples:

C/48N050W/M082F290F350
C/48N050W/M082F290PLUS
C/52N050W/M220F580F620

ITEM 16: DESTINATION AERODYNAMIC AND TOTAL ESTIMATED ELAPSED TME, ALTERNATE AERODROMES(S)
--

Destination aerodrome and total estimated elapsed time (8 characters)

INSERT the ICAO four letter location indicator of the destination aerodrome followed, without a space by the total estimated elapsed time,

OR if no location indicator has been assigned,

INSERT ZZZZ followed, without a space, by the total estimated elapsed time, and SPECIFY in Item 18 the name of the aerodrome, preceded by DEST/.

Note – For a flight plan received from an aircraft in flight, the total estimated elapsed time is the Estimated time from the first point of the route to which the flight plan applies.

Alternate aerodrome(s) (4 characters)

INSERT the ICAO four-letter location indicator(s) of not more than two alternate aerodromes,
Separated by a space,

OR, if no location indicator has been assigned to the alternate aerodrome,

INSERT ZZZZ and SPECIFY in Item 18 the name of the aerodrome, preceded by ALTN/.

ITEM 18: OTHER INFORMATION

INSERT	0 (zero) if no other information,
OR,	any other necessary information in the preferred sequence shown here under, in the form of the appropriate indicator followed by an oblique stroke and the information to be recorded:
EST/	Significant points or FIR boundary designators and accumulated estimated elapsed times to such points or FIR boundaries, when so prescribed on the basis of regional air navigation agreements, or by the appropriate ATS authority.
Examples:	EET/CAP 0745 XYZ0830 EET/EINN0204
RIF/	The route details to the revised destination aerodrome, followed by the ICAO four-letter location indicator of the aerodrome. The revised route is subject to re-clearance in flight.
Examples:	RIF/DTA HEC KLAX RIF/ESP G94 CLA APPH RIF/LEMD
REG/	The registration markings of the aircraft, if different from the aircraft identification in Item 7.
SEL/	SELCAL Code, if so prescribed by the appropriate ATS authority.
OPR/	Name of the operator, if not obvious from the aircraft identification in Item 7.
STS/	Reason for special handling by ATS, e.g. hospital aircraft, one engine inoperative, e.g. STS/HOSP, STS/ONE ENG INOP.
TYP/	Type(s) of aircraft, preceded if necessary by number(s) of aircraft, if ZZZZ is inserted in Item 9.
PER/	Aircraft performance data, if so prescribed by the appropriate ATS authority.
COM/	Significant data related to communication equipment as required by the appropriate ATS authority, e.g. COM/UHF only.
DAT/	Significant data related to data link capability, using one or more of the letters, S, H, V, and M, e.g. DAT/S for satellite data link, DAT/H for HF data link, DAT/V for VHF data link,
DAT/M for SSR Mode S data link.	
NAV	Significant data related to navigation equipment as required by the appropriate ATS authority, e.g. NAV/INS.
DEP/	Name of departure aerodrome, if ZZZZ is inserted in Item 13, or the ICAO four-letter location indicator of the location of the ATS unit from which supplementary flight plan data can be obtained, if AFIL is inserted in Item 13.
DEST/	Name of destination aerodrome, if ZZZZ is inserted in Item 16.
ALTN/	Name of destination alternate aerodrome(s), if ZZZZ is inserted in Item 16.
RALT/	Name of en route alternate aerodrome(s)
RMK/	Any other plan language remarks when required by the appropriate ATS authority or deemed authority.

ITEM 19: SUPPLEMENTARY INFORMATION

Endurance

After E/ INSERT a four-figure group giving the fuel endurance in hours and minutes.

Persons on board

After P/ INSERT the total number of persons (passengers and crew) on board, when required by the appropriate ATS authority. INSERT TBN (to be notified) if the total number of persons is not known at the time of filing.

Emergency and survival equipment

R/(RADIO) CROSS OUTU if UHF on frequency 243.0 MHz is not available. CROSS OUT V if VHF on frequency 121.5 MHz is not available. CROSS OUTE if emergency location beacon-aircraft (ELBA) is not available.

S/ CROSS OUT all indicators if survival equipment is not carried, CROSS OUTP if polar (SURVIVAL survival equipment is not carried. CROSS OUTD if desert survival equipment is not EQUIPMENT) carried. CROSS OUTM if maritime survival equipment is not carried. CROSS OUTJ if Jungle survival equipment is not carried.

J/(JACKETS) CROSS OUT all indicators if life jackets are not carried. CROSS OUTL if life jackets are not Equipped with lights. CROSS OUTF if life jackets are not equipped with fluorescent. CROSS OUTU or V or both as in R/ above to indicate radio capability of jackets, if any.

D/ CROSS OUT indicators D and C if no dinghies are carried, or INSERT number of dinghies (DINGHIES) carried; and (NUMBER)

(CAPACITY) INSERT total capacity, in persons, of all dinghies carried; and

(COVER) CROSS OUT indicator C if dinghies are not covered; and

(COLOR) INSERT colour of dinghies, if carried

A/ INSERT colour of aircraft and significant markings. (AIRCRAFT COLOR AND MARKINGS)

N/ CROSS OUT indicator N if no remarks, or INDICATE any other survival equipment carried (REMARKS) and any other remarks regarding survival equipment.

C/ (PILOT) INSERT name of pilot-in-command.

2.3 Filed by

INSERT the name of the unit, agency, or person filing the flight plan.

A flight plan shall be submitted prior to operating.

- a) Any flight to be provided with air traffic control services.
- b) Any IFR flight within advisory airspace.

Flight plans for intended flights shall be submitted to the appropriate ATS unit at least sixty (60) minutes before departure.

In the event of a proposed delay of 30 minutes or more the pilot-in-command or his representative will be required to notify ATC of the revised ETD. Where applicable the flight plan should be amended or a new flight plan submitted.

Co-ordinated Universal Time (UTC) shall be expressed in hours and minutes to ensure correct time to within plus or minus 30 seconds of UTC.

Wherever time is utilized in the application of data link communications, it shall be accurate to within 1 second of UTC.

Adherence to Flight Plan

An aircraft shall adhere to the current flight plan submitted for a controlled flight.

Deviations from the current flight plan

- a) Deviation from track- If the aircraft is off track, action shall be taken forthwith to adjust the heading of the aircraft to regain track as soon as practicable.
- b) Deviation from ATC assigned Mach number/indicated airspeed- The appropriate air traffic services unit shall be informed immediately.
- c) Deviation from Mach number/true airspeed- If the sustained Mach number/true airspeed at cruising level varies by plus or minus Mach 0.02 or more, or plus or minus 19 km (10kt) true airspeed or more from the current flight plan, the appropriate air traffic services unit shall be so informed.
- d) Change in time estimate- Except where ADS-C is activated and serviceable in airspace, if the time estimate for the next applicable reporting point, flight information region boundary or destination aerodrome, change in excess of two minutes from that notified to air traffic services, the flight crew shall notify the appropriate air traffic services unit as soon as possible.

POSITION REPORTS

Controlled flight shall report to the appropriate air traffic services unit, as soon as possible, the time and level of passing each designated compulsory reporting point.

COMMUNICATIONS

An aircraft operated as a controlled flight shall maintain continuous air-ground voice communication watch on the appropriate communication channel and establish two-way communication as necessary with the appropriate air traffic control unit.

SELCAL or similar automatic signalling devices satisfy the requirement to maintain an air-ground voice communication watch.

The requirement for an aircraft to maintain air-ground voice communication watch remains in effect after CPDLC has been established.

Communication failure: The aircraft shall attempt to establish communications with the appropriate air traffic control unit using all other available means. In addition, the aircraft, when forming part of the aerodrome traffic at a controlled aerodrome, shall keep a watch for such instructions as may be issued by visual signals.

If in instrument meteorological conditions:

- a) In airspace where radar is not used in the provision of air traffic control, maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 20 minutes following the aircraft's failure to report its position over a compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan.
- b) In airspace where radar is used in the provision maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 7 minutes following:
 - 1) The time the last assigned level or minimum flight altitude is reached
 - 2) The time the transponder is set to Code 7600
 - 3) The aircraft's failure to report its position over a compulsory reporting point, whichever is later, and thereafter adjust level and speed in accordance with the field flight plan.
- c) Commence descent from the navigation aid or fix as close as possible to, the expected approach time last received and acknowledged or if no expected approach time has been received and acknowledged, at or as close as possible to, the estimated time of arrival resulting from the current flight plan.
- d) Land, if possible, within thirty minutes after the estimated time of arrival or the last acknowledged expected approach time, whichever is later.

VISUAL FLIGHT RULES

Except when operating as a special VFR flight, VFR flights shall be conducted so that the aircraft is flown in conditions of visibility and distance from clouds equal to or greater than those specified in Table.

Except when a clearance is obtained from an air traffic control unit flights shall not take off or land at an aerodrome within a control zone, or enter the aerodrome traffic zone or traffic pattern:

- a) When the ceiling is less than 450m (1500ft)
- b) When the ground visibility is less than 5 km

VFR flights shall be operated during the period from 20 minutes before sunrise to 20 minutes after sunset, except when exempted by air traffic control for local flights and such training flights of flying club aircraft as may be cleared by air traffic control. Local flight is a flight wholly conducted in the immediate vicinity of an aerodrome.

VFR flights shall not be operated-

- a) Above FL150
- b) At transonic and supersonic speeds
- c) More than 100NM seaward from the shoreline within controlled airspace.

VFR flight shall not be flown:

- a) Over the congested areas of cities, towns or settlements or over an open air assembly of persons at a height less than 300m (1000ft) above the highest obstacle within a radius of 600 m from the aircraft.
- b) Elsewhere, at a height less than 150 m (500ft) above the ground or water.

VFR flights in level cruising flight when operated above 900m (3000ft) from the ground or water, or a higher datum as specified by the appropriate ATS authority, shall be conducted at a cruising level appropriate to the track as specified in the Tables of cruising levels.

VFR flights shall comply with the provision when operated within Class D airspace.

SPECIAL VFR FLIGHTS

- a) When traffic conditions permit, special VFR flights may be authorised subject to the approval from the unit providing approach control service.
- b) When the ground visibility is not less than 1500 meters, special VFR flights may be authorised to enter a control zone for the purpose of landing to take-off and depart from a control zone, to cross a control zone, or to operate locally within a control zone.
- c) IMC visibility and distance from cloud minima

Table 3-1* (see 4.1)

Altitude band	Airspace class	Flight visibility	Distance from cloud
At and above 3 050 m (10 000 ft) AMSL	A*** B C D E F G	8Km	1 500 m horizontally 300 m (1 000 ft) vertically
Below 3 050 m (10 000 ft) AMSL and above 900 m (3 000 ft) AMSL, or above 300 m (1 000 ft) above terrain, whichever is the higher	A*** B C D E F G	5Km	1 500 m horizontally 300 m (1 000 ft) vertically
At and below 900 m (3 000 ft) AMSL, or 300 m (1 000 ft) above terrain, whichever is the higher	A***B C D E	5Km	1 500 m horizontally 300 m (1 000 ft) vertically
	F G	5Km**	Clear of cloud and with the surface in sight.

* When the height of the transition altitude is lower than 3 050 m (10 000 ft) AMSL, FL 100 should be used in lieu of 10 000 ft.

** When so prescribed by the appropriate ATS authority:

a) flight visibilities reduced to not less than 3 000 m may be permitted for flights operating:

- 1) at speeds that, in the prevailing visibility, will give adequate opportunity to observe other traffic or any obstacles in time to avoid collision; or
- 2) in circumstances in which the probability of encounters with other traffic would normally be low, e.g. in areas of low volume traffic and for aerial work at low levels.

b) HELICOPTERS may be permitted to operate upto 1 000 m flight visibility, if manoeuvred at a speed that will give adequate opportunity to observe other traffic or any obstacles in time to avoid collision.

***The VMC minima in Class A airspace are included for guidance to pilots and do not imply acceptance of VFR flights in Class A airspace.

INSTRUMENT FLIGHT RULES

RULES APPLICABLE TO ALL IFR FLIGHTS

Aircraft shall be equipped with suitable instruments and with navigation equipment appropriate to the route to be flown.

Minimum Levels

- a) Over high terrain or in mountainous areas, at a level which is at least 600m (2000ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.
- b) Elsewhere which is at least 300m (1000ft) above the highest obstacle located within 8km of the estimated position of the aircraft.

RULES APPLICABLE TO IFR FLIGHTS OUTSIDE CONTROLLED AIRSPACE

Communications-

An IFR flight operated outside controlled airspace but within or into areas or along routes, designated by the appropriate ATS authority shall maintain an air ground voice communication watch on the appropriate communication channel and establish two-way communication as necessary.

Flight by Night-

A flight performed between half an hour after sunset and half an hour before sunrise.

DISTRESS AND URGENCY SIGNALS

Distress Signals

- a) A radiotelephony distress signal consisting of the spoken word MAYDAY
- b) A distress message sent via data link which transmits the intent of the word MAYDAY
- c) Rockets or shells throwing red lights, fired one at a time at a short interval
- d) A parachute flare showing a red light.

Urgency Signals

The following signals, used either together or separately mean that an aircraft wishes to give notice of difficulties which compel it to land without requiring immediate assistance.

- a) The repeated switching on and off of the landing lights
- b) The repeated switching on and off of the navigation lights.

The following signals mean that an aircraft has a very urgent message to transmit concerning the safety of a ship, aircraft.

- a) Consisting of the group XXX
- b) A radiotelephony urgency signal consisting of the spoken words PAN, PAN
- c) An urgency message sent via data link which transmits the intent of the words PAN, PAN

Procedures for the issue of Air Defence Clearance (ADC):

Except the local flights conducted within airspace of 5NM radius centred at ARP and vertical limits of 1000ft.AGL of an aerodrome.

- All flights shall obtain Air Defence Clearance before entering ADIZ from respective FIC ten minutes prior to entering Indian Airspace.
- ADC shall be obtained before departure and in the event of departure being delayed for more than 60 minutes. A fresh ADC shall be obtained.

VISUAL SIGNALS USED TO WARN IN OR ABOUT TO ENTER A RESTRICTED, PROHIBITED OR DANGER AREA

By day and by night a series of projectiles discharged from the ground at intervals of 10 seconds, each showing, on bursting, red and green lights or stars will indicate to an unauthorized aircraft that it is flying in or about to enter a restricted, prohibited or danger area.

SIGNALS FOR AERODROME TRAFFIC

LIGHTS AND PYROTECHNIC SIGNALS

Instructions from Aerodrome Control Tower to Aircrafts

Direct	Light towards ac concerned	Ac in flight	Ac on the ground
Directed towards aircraft concerned	Steady green 	Clear to land	Clear for take-off
	Steady red 	Give way to other ac and continue circling	Stop
	Series of green flashes 	Return for landing and await landing clearance	Cleared for taxi
	Series of red flashes 	Aerodrome unsafe, do not land or complete refusal to land	Taxi clear of landing area in use
	Series of white flashes 	Landing at this aerodrome after receiving clearance to land and then proceed to the apron	Return to the starting point on the aerodrome
	Red pyrotechnics light 	Notwithstanding any previous instruction, do not land for the time being	

ACKNOWLEDGEMENT BY AN AIRCRAFT

A) When in flight:

- (i) During the hours of daylight: by rocking the aircrafts wings
- (ii) During the hours of darkness: by flashing on and off twice the aircrafts landing lights or if not so equipped, by switching on and off twice its navigation lights.

B) When on the ground:

- (i) During the hours of daylight : by moving the aircrafts ailerons or rudder
- (ii) During the hours of darkness : by flashing on and off twice the aircrafts landing lights or if not so equipped by switching on and off twice its navigation lights.

VISUAL GROUND SIGNALS

PROHIBITION OF LANDING:

A horizontal red square panel with yellow diagonals when displayed in a signal area indicates that landings are prohibited and that the prohibition is liable to be prolonged.

NEED FOR SPECIAL PRECAUTIONS WHILE APPROACHING OR LANDING:

A horizontal red square panel with one yellow diagonal when displayed in a signal area indicates that owing to the bad state of the manoeuvring area, or for any other reason, special precautions must be observed in approaching to land or in landing.

USE OF RUNWAYS AND TAXIWAYS

A horizontal white dumbbell when displayed in a signal area indicates that aircraft are required to land, take off and taxi on runways and taxiways only.

The same horizontal white dumbbell but with a black bar placed perpendicular to the shaft across each circular portion of the dumbbell when displayed in a signal area indicates that aircraft are required to land and take off on runways only, but other manoeuvres need not be confined to runways and taxiways.

AIR TRAFFIC SERVICES REPORTING OFFICE:

The letter C displayed vertically in black against a yellow background indicates the location of the air traffic services reporting office.

GLIDER FLIGHTS IN OPERATION:

A double white cross displayed horizontally in the signal area indicates that the aerodrome is being used by gliders and that glider flights are being performed.

MARSHALLING SIGNALS

FROM A SIGNALMAN TO AN AIRCRAFT

Normal Stop: Fully extended arms and wands at a 90-degree angle to sides and slowly move to above head until wands cross.

Emergency Stop: Abruptly extend arms and wands to top of head crossing wands.

Chocks Inserted: With arms and wands fully extended above head, move wands inward in a jabbing motion until wand touch. Ensure and acknowledgement is received from flight crew.

Chocks Removed: With arms and wands fully extended above head, move wands outward in a jabbing motion. Do not remove chocks until authorised by the flight crew.

Affirmative/all clear: Raise right arm to head level with wand pointing up or display hand with THUMBS UP left arm remains at side by knee.

FROM THE PILOT OF AN AIRCRAFT TO A SIGNALMAN

BRAKES

- a) Brakes engaged: raise arm and hand, with fingers extended, horizontally in front of face, then clench fist.
- b) Brakes released: raise arm, with fist clenched, horizontally in front of face, then extend fingers.

CHOCKS

- a) Insert chocks: arms extended, palms outwards, move hands inwards to cross in front of face.
- b) Remove chocks: hands crossed in front of face, palms outwards, move arms outwards.

AIR TRAFFIC SERVICES

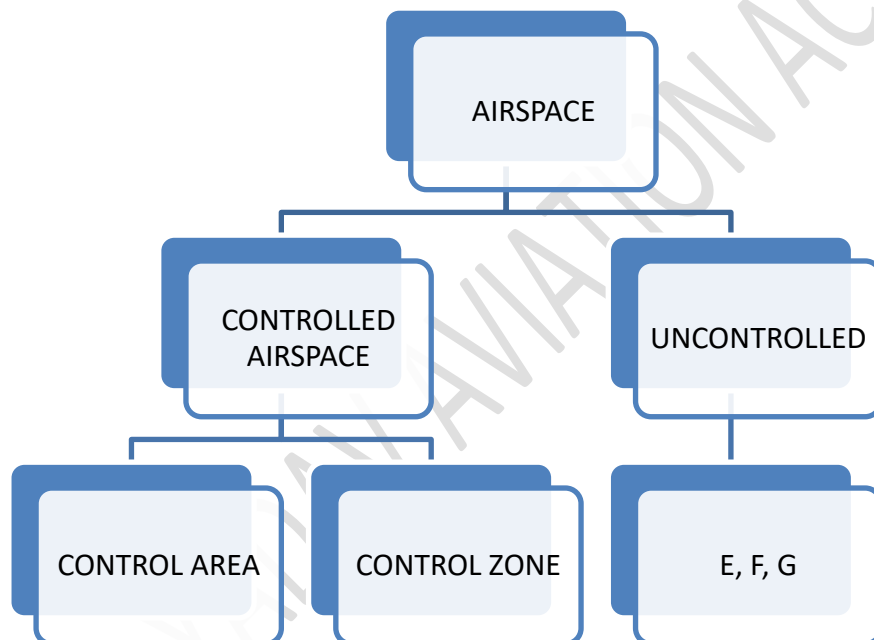
(Ref:Annex 11,Doc.4444 and CAR SECTION 9 – AIR SPACE AND AIR TRAFFIC MANAGEMENT SERIES E PART I ISSUE II)

GENERAL

Objectives of the Air traffic Services:

- a) Prevent collisions between aircraft
- b) Prevent collisions between aircraft on the manoeuvring area
- c) Expedite and maintain an orderly flow of air traffic
- d) Notify appropriate organizations regarding aircraft in need of search and rescue aid.

CLASSIFICATION OF AIRSPACES:



Class A: IFR flights only are permitted, all flights are provided with air traffic control service and are separated from each other.

Class B: IFR and VFR flights are permitted, all flights are provided with air traffic control service and are separated from each other.

Class C: IFR and VFR flights are permitted, all flights are provided with air traffic control services and IFR flights are separated from other IFR flights and from VFR flights. VFR flights are separated from IFR flights and receive traffic information in respect of the other VFR flights.

Class D: IFR and VFR flights are permitted and all flights are provided with air traffic control service. IFR flights are separated from other IFR flights and receive traffic information in respect of VFR flights, VFR flights receive traffic information in respect of all other flights.

Class E: IFR and VFR flights are permitted, IFR flights are provided with air traffic control service and are separated from other IFR flights. All flights receive traffic information as far as is practical. Class E shall not be used for control zones.

Class F: IFR and VFR flights are permitted, all participating IFR flights receive an air traffic advisory service and all flights receive flight information service if requested.

Class G: IFR and VFR flights are permitted and receive flight information service if requested. Airspace classes have been classified in AIP India ENR 1.4.

Class	Rules	Service	Clearance	Separation	FIS	Speed Limit (NOTE 1)	Radio Watch	CTR SVFR	Use
A	IFR	ATC	YES	ALL		NO	YES	YES	CTA: CTR AIRWAYS: ROUTES
B	IFR VFR	ATC	YES	ALL		NO	YES	YES	CTA: CTR AIRWAYS: ROUTES
C	IFR	ATC	YES	IFR-IFR		NO	YES	YES	CTA: CTR (AIRWAYS) (NOTE 5)
		IFR-VFR		IFR-VFR					
	VFR	ATC FIS		VFR-IFR VFR-VFR		YES			
D	IFR	ATC	YES	IFR-IFR		YES	YES	YES	CTA: CTR (AIRWAYS) (NOTE 5)
		IFR-VFR		IFR-VFR					
	VFR	FIS (NOTE 4)		VFR-IFR VFR-VFR					
E	IFR	ATC	YES	IFR-IFR		YES	YES	NO (NOTE 2)	CTA: (AIRWAYS) (NOTE 5)
		IFR-VFR	NO	IFR-VFR					
	VFR	FIS		VFR-IFR VFR-VFR			NO		
F	IFR	ADV ATC	NO	IFR-IFR (NOTE 3)	ALL	YES	YES	NO	ADVISORY ROUTES
	IFR						YES		
	VFR	FIS					NO		
G	IFR	FIS	NO	NONE	ALL	YES	YES	NO	OPEN FIR; FIS ROUTES
	VFR						NO		

Note 1: General Speed Limit is 250 kts IAS enforced below FL100 (10 000ft)

Note 2: Separation as far as is practical

Note 3: Service to participating IFR traffic only (non participating IFR to file FP and maintain 2-way comms)

Note 4: Radar information/advice may be provided to augment FIS in CAS

Note 5: An airway will adopt the classification of the surrounding airspace when it passes through a CTA

ADV → Adv

Prohibited Areas:

Air airspace of defined dimensions above the land areas or territorial waters of India within which the flights are not permitted at any time under any circumstances. VA(P)-2, indicates Prohibited Area No.2 in Mumbai FIR.

Restricted Areas:

An airspace of defined dimensions above the land areas or territorial waters of India within which the flight of aircraft is restricted(R)-81, is Restricted Area No. 81 in Chennai FIR.

Danger Areas:

An airspace of defined dimensions within which activities dangerous to the flight of aircraft exist at the specified times. Such times are notified by NOTAMA's.

LOCAL / TRAINING FLIGHTS

For local training or test flights conducted by scheduled/non-scheduled operators, a flight plan shall be submitted as for non-scheduled flights.

For training flights conducted by flying clubs within aerodrome traffic zone, a flight plan may be submitted on telephone.

Note: - On aerodrome where no aerodrome traffic zone is established or no airspace has been reserved for local flying, on area of 5 NM radius aerodrome reference point and vertical limits up to 3000ft AGL shall be treated as the vicinity of aerodrome for VFR flights.

Flight plan for cross-country flights conducted by flying club aircraft and general aviation aircraft from airfields where ATS reporting office does not exist, shall be submitted to the FIC on phone, giving the information.

A lower limit of a control area shall be established at a height above the ground or water of not less than 200m (700 ft)

FLIGHT INFORMATION REGIONS OR CONTROL AREAS IN THE UPPER AIRSPACE

Control Zones – The lateral limits of control zones shall encompass at least those portions of the airspace, which are not within control areas, containing the paths of IFR flights arriving at and departing from aerodromes to be used under instrument meteorological conditions.

The lateral limits of a control zone shall extend to at least 9.3 km (5NM) from the centre of the aerodrome or aerodromes concerned in the directions from which approaches.

Service to aircraft in the event of an emergency: - An aircraft known or believed to be in a state of emergency, including being subjected to unlawful interference, shall be given maximum consideration, assistance and priority over other aircrafts as may be necessitated by the circumstances.

ALERTING SERVICE AND EMERGENCY PHASES OF FLIGHT

Alerting service shall be provided –

- a) For all aircraft provided with air traffic control service.
- b) In so far as practicable, to all other aircraft having filed plan or otherwise known to the air traffic services.
- c) To any aircraft known or believed to be the subject of unlawful interference.

a) Uncertainty phase (INCERFA) when:

- 1) no communication has been received from an aircraft within a period of thirty minutes after the time a communication should have been received or from the time an unsuccessful attempt to establish communication with such aircraft was first made, whichever is the earlier, or when
- 2) an aircraft fails to arrive within thirty minutes of the estimated time of arrival last notified to or estimated by air traffic services units, whichever is the later, when no doubt exists as to the safety of the aircraft and its occupants.

b) Alert Phase (ALERFA) when:

- 1) following the uncertainty phase, subsequent attempts to establish communication with the aircraft or inquiries to other relevant sources have failed to reveal any news of the aircraft, or when.
- 2) an aircraft has been cleared to land and fails to land within five minutes of the estimated time of landing and communication has not been re-established with the aircraft, or when.
- 3) information has been received which indicates that the operating efficiency of the aircraft has been impaired, but not to the extent that a forced landing is likely, except when evidence exists that would allay apprehension as to the safety of the aircraft and its occupants, or when.
- 4) an aircraft is known or believed to be the subject of unlawful interference.

c) Distress Phase (DETRESFA) when:

- 1) following the alert phase, further unsuccessful attempts to establish communication with the aircraft and more widespread unsuccessful inquiries point to the probability that the aircraft is in distress or when.
- 2) the fuel on board is considered to be exhausted or to be insufficient to enable the aircraft to reach safety, or when.
- 3) information is received which indicates that the operating efficiency of the aircraft has been impaired to the extent that a forced landing is likely or when.
- 4) information is received or it is reasonably certain that the aircraft is about to make or has made a forced landing except when there is reasonable certainty that the aircraft and its occupants are not threatened by grave and imminent danger and do not require immediate assistance.

PRINCIPLES GOVERNING THE IDENTIFICATION OF RNP TYPES AND THE IDENTIFICATION OF ATS ROUTES OTHER THAN STANDARD DEPARTURE AND ARRIVAL ROUTES

Composition of Designator

Selection of the letter shall be made from those listed hereunder –

- a) A, B, G, R for routes which form part of the regional networks of ATS routes and are not area navigation routes.
- b) L, M, N, P for area navigation routes which form part of the regional networks of ATS routes.
- c) H, J, V, W for routes which do not form part of the regional networks of ATS routes and are not area navigation routes.
- d) Q, T, Y, Z for area navigation routes which do not form part of the regional networks of ATS routes.
- 1) K to indicate a low-level route established for use primarily by helicopters.
- 2) U to indicate that the route or portion thereof is established in the upper airspace.
- 3) S to indicate a route established exclusively for use by supersonic aircraft during acceleration, deceleration and while in supersonic flight.
- a) The letter F to indicate that on the route or portion thereof advisory service only is provided.
- b) The letter G to indicate that on the route or portion thereof flight information service only is provided.

ATS Routes in India

- Aircraft shall obtain ATC clearance from the appropriate ATS unit at least 10 minutes prior to entering controlled airspace.
- All aircraft are forbidden to operate within 15NM of the international border of India unless specifically permitted or except when following the ATS route or operating to and from any aerodrome situated within 15NM of the international border of India.
- ATS routes with W designators are exclusively for domestic operators only.

Separation of Traffic

AREA CONTROL SERVICE

(PANS ATM Doc 444, AIP, India)

VERTICAL SEPARATION

Vertical separation Minimum

- a) A nominal 1000 feet below FL290 and a nominal 2000 feet at or above FL290, except as provided.
- b) A nominal 1000 feet when both aircraft are RVSM compliant and operating within designated RVSM airspace.

Horizontal Separation

- a) Lateral separation
 - b) Longitudinal separation
 - c) Radar separation
-
- a) **VOR** - Both aircraft are established on radials diverging by at least 15 degrees and at least one aircraft is 15NM or more from the facility.
 - b) **NDB** - Both aircraft are established on tracks to or from the NDB which are diverging by at least 30 degrees and at least one aircraft is 15 NM or more from the facility.
 - c) **DR (dead reckoning)** - Both aircraft are established in tracks diverging by at least 45 degrees and at least one aircraft is 15 NM or more from the point of intersection of the tracks.

Aircraft at the same cruising level

Aircraft flying on the same track

- a) 15 minutes
- b) 10 minutes if navigation aids permit frequent determination of position and speed.
- c) 5 minutes in the following cases, provided that in each case preceding aircraft is maintaining a true airspeed faster than the succeeding aircraft of 37km/h (20 Knots) or more.
- d) 3 minutes in the cases listed under (c) provided that in each case the preceding aircraft is maintaining a true airspeed faster than the succeeding aircraft of 74 km/h (40 knots) or more.

Longitudinal separation minima based on using DME and/or GNSS

Aircraft on the same track

- a) 20 NM, provided:
 - 1) Each aircraft utilizes-
 - i) The same on track DME station when both aircraft are utilizing DME
 - ii) An on track DME station and a collocated waypoint when one aircraft is utilizing DME and the other is utilizing GNSS
 - iii) The same waypoint when both aircraft are utilizing GNSS
- b) 10 NM provided
 - 1) The leading aircraft maintains a true airspeed of 20 kts or faster than the succeeding aircraft.
 - 2) Each aircraft utilizes-
 - i) The same on track DME station when both aircraft are utilizing DME
 - ii) An on track DME station and a collocated way point when one aircraft is utilizing DME and the other is utilizing GNSS
 - iii) The same waypoint when both aircraft are utilizing GNSS

SUMMARY AND LONGITUDINAL SEPARATION

APPROACH CONTROL SERVICE

(Doc 4444, AIP, India)

Minimum Separation between Departing Aircraft

One-minute separation if aircraft are to fly on tracks diverging by at least 45 degrees immediately after take-off so that lateral separation is provided.

Two minutes between take-offs when the preceding aircraft is 40 kt or faster than the following aircraft and both aircraft propose to follow the same track.

Five minutes separation while vertical separation does not exist if a departing aircraft will be flown through the level of a preceding departing aircraft and both aircraft propose to follow the same track. Action must be taken to ensure that the five-minute separation will be maintained or increased while vertical separation does not exist.

PROCEDURES FOR ARRIVING AIRCRAFT

Visual Approach

- Visual approach is an approach by an IFR flight when either part or all of an instrument approach procedure is not completed and the approach is executed in visual reference to terrain.
- Clearance for an IFR flight to execute a visual approach may be requested by the pilot or initiated by the controller.
- Separation shall be provided between an aircraft cleared to execute a visual approach and other arriving and departing aircraft.
- When clearance to execute visual approach has been issued it shall be the responsibility of pilot to maintain terrain clearance.

Visual approach by a flight crew:

A flight crew may request visual approach if he has runway in sight and the pilot can maintain visual reference to terrain.

Expected Approach Time (EAT)

An expected approach time shall be determined for an arriving aircraft that will be subjected to a delay of 10 minutes or more. The expected approach time shall be transmitted to the aircraft as soon as practicable and preferably not later than at the commencement of its initial descent from cruising level. A revised expected approach time shall be transmitted to the aircraft without delay whenever it differs from that previously transmitted by 5 minutes or more, or such lesser period of time as agreed between the ATS units concerned.

AERODROME CONTROL SERVICE

(Doc 4444, AIP, INDIA)

FUNCTIONS OF AERODROME CONTROL TOWERS

Aerodrome control towers shall issue information and clearances to aircraft under their control to achieve a safe, orderly and expeditious flow of air traffic on and in the vicinity of an aerodrome with the object of preventing collisions.

- Aerodrome controller- normally responsible for operations on the runway and aircraft flying within the area of responsibility of the aerodrome control tower.
- Ground controller- normally responsible for traffic on the manoeuvring area with the exception of runways.
- Clearance delivery position- normally responsible for delivery of start-up and ATC clearance to departing IFR flights.

ALERTING SERVIC PROVIDED BY AERODROME CONTROL TOWERS

Aerodrome control towers are responsible for alerting the rescue and firefighting services whenever-

- a) An aircraft accident has occurred on or in the vicinity of the aerodrome.
- b) Information is received that the safety of an aircraft which is or will come under the jurisdiction of the aerodrome control tower may have or has been impaired.

SUSPENSION OF VISUAL FLIGHT OPERATIONS

- A) The approach control unit or the appropriate ACC
- B) The aerodrome control tower.

Essential Local Traffic Information

Essential local traffic shall be considered to consist of any aircraft, vehicle or personnel on or near the manoeuvring area or traffic operating in the vicinity of the aerodrome, which may constitute a hazard to the aircraft concerned.

Runway Incursion

Runway incursion by an aircraft or vehicle, or animal or flocks of birds on the runway.

Wake Turbulence

Aerodrome controllers shall when applicable apply the wake turbulence separation minima specified. Whenever the responsibility for wake turbulence avoidance rests with the pilot-in command, aerodrome controllers shall to the extent practicable, advise aircraft of the expected occurrence of hazards caused by turbulent wake.

CONTROL OF OTHER THAN AIRCRAFT TRAFFIC

Entry to the manoeuvring area

The movement of persons or vehicles including towed aircraft on the manoeuvring area of an aerodrome shall be controlled by the aerodrome control tower.

AERONAUTICAL GROUND LIGHTS

The lights of a visual approach slope indicator system shall be operated during the hours of daylight as well as of darkness and irrespective of the visibility conditions when the associated runway is being used.

Runway Lighting

Runway lighting includes such lights as edge, threshold, centre line, end, touchdown zone and wing bar lights.

Designation of Hot Spot

The aerodrome operator shall whenever necessary designate a location or several locations on the movement area of the aerodrome as hot spot.

USE OF AIR TRAFFIC SERVICES SURVEILLANCE SYSTEM

(DOC 4444 AND AIP, INDIA)

Procedural separation minima will be applied –

- i) In the event of ATS surveillance service failure.
- ii) In the area outside the ATS surveillance service coverage.
- iii) To aircraft leaving ATS surveillance service coverage or entering adjacent FIR except where ATS surveillance service transfer is affected.

Separation minima based on ATS surveillance systems in India

The following horizontal radar separation minima shall be applied:

- a) 5 NM horizontal radar separation up to 60 NM from radar head.
- b) 10 NM horizontal radar separation beyond 60 NM from radar head.

Separation minima based on ATS surveillance systems (ICAO)

Unless otherwise prescribed the horizontal separation, minimum based on radar shall be 9.3 km (5.0 NM)

The radar separation minimum in May, if so prescribed by the appropriate ATS authority, be reduced but not below:

- a) 5.6 km (3.0 NM) when radar and/or ADS-B and/or MLAT capabilities at a given location so permit it.
- b) 4.6 km (2.5 NM) between succeeding aircraft which are established on the same final approach track within 18.5 km (10 NM) of the runway end threshold.

Speed Control procedures- All aircraft (including arrivals and departures) operating below 10000 ft will fly at IAS not greater than 250 Kt. All

arriving aircraft, operating below 10000 Ft within 15 NM radius of VOR/DME serving the aerodrome will fly at IAS not greater than 220KT.

Procedure for operation of SSR transponder codes- All aircraft carrying serviceable transponder shall operate the transponder at all time during flight within Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR regardless of whether the aircraft is within or outside airspace where SSR is used for ATS.

Operating Procedures –

- 1) Aircraft departing from an aerodrome located in Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR shall be assigned an appropriate SSR code on departure. This SSR code setting shall continue unless instructed otherwise.
- 2) Aircraft not assigned a SSR code shall operate transponder on mode A3 code 2000 before entry into Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR and maintain that code setting until otherwise instructed.
- 3) In order to avoid interference on radar display, the pilot shall not operate the transponder when the aircraft is on ground except when entering the runway for take-off or till vacating the runway after landing.

AERONAUTICAL INFORMATION SERVICES

(ICAO ANNEX 15, AIP, INDIA AND CIVIL AVIATION REQUIREMENTS SECTION 9 – AIR SPACE AND AIR TRAFFIC MANAGEMENT SERIES I PART I ISSUE II)

The object of the aeronautical information service (IAS) is to insure the flow of aeronautical data and aeronautical information necessary for global air traffic management (ATM) system safety, regularity, economy and efficiency in an environmentally sustainable manner.

Responsibilities and Functions –

Aeronautical information service (excluding publications of Aeronautical Information Circulars (AIC) shall be provided by Airports Authority of India (AAI). AIC shall be published by DGCA.

AERONAUTICAL INFORMATION PUBLICATIONS (AIP) AIP are intended primarily to satisfy international requirement for the exchange of aeronautical information of a lasting character essential to air navigation. When practicable, the form of presentation is designed to facilitate their use in flight.

AIP constitute the basic information source for permanent information and long duration temporary changes.

Specifications for AIP Supplements

- Temporary changes of long duration (three months or longer) and information of short duration which contains extensive text and/or graphics shall be published as AIP Supplements.
- A checklist of valid AIP Supplements shall be issued at intervals of not more than one month. This information shall be issued through the medium of the monthly printed language list of valid NOTAMs.

NOTAMS

NOTAM(N) – Notam new

NOTAM(R) – Notam replacement

NOTAM(C)- Cancellation of Notam

Origination

- A NOTAM shall be originated and issued promptly whenever the information to be distributed is of a temporary nature and of short duration or when operationally significant permanent changes, or temporary changes of long duration are made at short notice, except for extensive text and/or graphics.
- A NOTAM shall be originated and issued concerning the following information-
 - a) Establishment, closure or significant changes in operation aerodromes/heliports or runways.
 - b) Establishment, withdrawal and significant changes in operation of aeronautical services (AGA, AIS, ATS, COM, MET, SAR etc.)
 - c) Establishment, withdrawal and significant changes in operational capability of radio navigation and air-ground communication service. Change of frequencies, change in notified hours of service, change of identification, change of orientation, change of location.
 - d) Establishment, withdrawal or significant change made to visual aids.
 - e) Interruption of or return to operation of major components of aerodrome lighting systems.
 - f) Establishment, withdrawal or significant changes made to procedures for air navigation services.
 - g) Changes to and limitations on availability of fuel, oil, and oxygen.
 - h) Major changes to search and rescue facilities and services available.
 - i) Erecting or removal of or changes to obstacles to air navigation in the take-off/climb, missed approach, approach areas and runway strip.
- The following information shall not be notified by NOTAM:
 - a) Routine maintenance work on aprons and taxiways which does not affect the safe movement of aircraft.

- b) Runway marking work, when aircraft operations can safely be conducted on other available runways.
- c) Temporary obstructions in the vicinity of aerodromes/heliports that do not affect the safe operation of aircraft.
- d) Partial failure of aerodrome/heliport lighting facilities where such failure does not directly affect aircraft operations.
- e) Partial temporary failure of air-ground communications when suitable alternative frequencies are known to be available and are operative.
- At least seven days advance notice shall be given of the activation of establishment danger, restricted or prohibited areas and of activities requiring temporary airspace restrictions other than for emergency operations.
- NOTAM notifying unserviceability of aids to air navigation, facilities or communication services shall give an estimate of the period of unserviceability or the time at which restoration of service is expected.

General Specifications

- Information concerning snow, slush, ice and standing water on aerodrome/heliport pavements shall, when reported by means of a SNOWTAM, contain the information in the order shown in the SNOWTOM format.
- Information concerning an operationally significant change in volcanic activity, a volcanic eruption and/or volcanic ash cloud shall, when reported by means of an ASHTAM, contain the information in the order shown in the ASHTAM Format.
- A checklist of NOTAM shall refer to the latest AIP Amendments, AIP Supplements and at least the internationally distributed AIC.

AERONAUTICAL INFORMATION REGULATION AND CONTROL (AIRAC)

- Information concerning establishment, withdrawal or significant changes upon a series of common effective dates at intervals of 28 days.
- The regulated system (AIRAC) shall also be used for the provision of information relating to the establishment and withdrawal of and premeditated significant changes.

INFORMATION TO BE NOTIFIED BY AIRAC

PART I

The establishment, withdrawal of, and premeditated significant changes (including operational trials) to:

- Positions, frequencies, call signs known irregularities and maintenance periods of radio navigation aids and communication facilities.
- Holding and approach procedures, arrival and departure procedures, noise abatement procedures and other pertinent ATS procedures.
- Meteorological facilities (including broadcasts) and procedures.
- Runways and stop ways.

PART II

The establishment and withdrawal of, and premeditated significant changes to:

- Position, height and lighting of navigation obstacles.
- Taxiways and apron.
- Hours of service: aerodromes, facilities and services.
- Customs, immigration and health services.
- Temporary danger, prohibited and restricted areas and navigational hazards, military exercises and mass movement of aircraft.
- Whenever major changes are planned and where advance notice is desirable and practicable, information published in paper copy form should be distributed by the AIS unit at least 56 days in advance of the effective date.

AERONAUTICAL INFORMATION CIRCULARS (AIC)

- A) An AIC shall be originated whenever it is necessary to promulgate aeronautical information which does not qualify:
 - For inclusion in an AIP
 - For the origination of a NOTAM
- B) An AIC shall be originated-
 - A long-term forecast of any major change in legislation, regulations.
 - Information of a purely explanatory or advisory nature liable to affect flight safety.
 - Information or notification of an explanatory or advisory nature concerning technical, legislative or purely administrative matters

Advice on medical matters of special interest to pilots.

General Specifications

- Each AIC shall be allocated a serial number.
- A checklist of AIC currently in force shall be issued at least once a year.

PRE-FLIGHT AND POST-FLIGHT INFORMATION/DATA

Pre-flight information-

Current NOTAM and other information of urgent character shall be made available to flight crews in the form of plain-language pre-flight information bulletins (PIB).

NATIONAL PROVISIONS

INTERNATIONAL NOTAM OFFICE (NOF)

Four International NOTAM offices are established at Mumbai, Calcutta, Delhi and Chennai.

AREA OF RESPONSIBILITY

The Aeronautical Information Service is responsible for the collection and dissemination of information for the entire territory of India and for airspace over the high seas under the jurisdiction of India for air traffic control purpose.

AERONAUTICAL PUBLICATIONS

- a) Aeronautical Information Publication (AIP)
- b) AIP SUP
- c) NOTAM and pre-flight information bulletins (PIB)
- d) Check lists and summaries

Volume I contains Part I – General (Gen) and Part 2 – En-route(ENR) Volume II contains Part 3 – Aerodromes (AD). These parts contain following information.

AERONAUTICAL INFORMATION PUBLICATIONS SHALL INCLUDE IN PART I – GENERAL (GEN)

- National Regulation and Requirements
- Tables and codes
- Services
- Air traffic services
- Communication services

- Meteorological services
- Search and rescue
- Charges for aerodromes/heliports and Air Navigation Services

AERONAUTICAL INFORMATION PUBLICATIONS SHALL INCLUDE IN PART 2 EN-ROUTE (ENR)

- General Rules and Procedures.
- FIR/CTA/TMA, Other Regulated Airspace
- Radio Navigation Aids/systems
- Navigation warnings: Prohibited, Restricted and Danger Areas

PART 3 AERODROME (AD)

NOTAM and Pre-flight Information Bulletins (PIB)

NOTAMs are originated by the International NOTAM Office Mumbai, Kolkata, Delhi, and Chennai. NOTAMs are distributed in five series identified by the letters A, B, C, D and G as follows-

Series A: Contain information in respect of Changes / unserviceability etc. of aeronautical facilities, likely to last for more than 2 hours, in respect of locations of direct importance to International aircraft operations.

Series B: Contain information in respect of Changes/ unserviceability etc. Of aeronautical facilities, likely to last for more than 30 minutes but less than 2 hours in respect of locations of direct importance to International aircraft operations.

Series C: Contain information in respect of changes/unserviceability etc. Of aeronautical facilities in respect of locations utilized by domestic flights only and for which no separate A series International distribution is given.

Series D: Contain information in respect of changes/unserviceability etc. Of aeronautical facilities in respect of locations of military controlled airfields utilized by domestic flights the information of which is issued by AHQ through a VVO broadcast.

Series G: Contain information of general and lasting character affecting aircraft operations in general. The series is operated only by the International NOTAM Office, Delhi and issued under the authority of the NOTAM of each series are allocated.

Checklist and Summary of NOTAM

A checklist of all NOTAM current on the 1st of each month is originated by the respective NOTAM office and transmitted over the AFTN.

The complete summary is prepared once a year on the 1st of January, but the summary for the subsequent months shall contain the text of only those NOTAM which have been promulgated during the month of issue.

SEARCH AND RESCUE

Search and Rescue Services

Contracting States shall arrange for the establishment and prompt provision of search and rescue services within their territories. Such services shall be provided on a 24-hour basis.

Those portions of the high seas or areas of undetermined sovereignty for which search and rescue services will be established shall be determined on the basis of Regional Air Navigation Agreements.

NATIONAL PROVISIONS

Search and rescue services are provided over land areas by the National Aeronautical Search & Rescue coordination committee (NASARCC) designating Secretary, Ministry of Civil Aviation as its chairman & over oceanic areas by National Maritime SAR Board (NMSARB) designating Director General Indian Coast Guard as its Chairman.

Airports Authority of India coordinates SAR services through 4 rescue Coordination Centres established at Delhi, Mumbai, Kolkata & Chennai FIRs.

In areas over high seas, SAR services are provided by Indian Coast Guard through 3 maritime Rescue coordination centres (MRCCs) located at Mumbai, Chennai & Port Blair. And Rescue services over Indian Airspace excluding the oceanic areas are coordinated by Airport Authority of India with the SAR agencies.

Search and Rescue services over oceanic areas which form part of Kolkata, Mumbai & Chennai FIR are provided by Indian Coast Guard.

OPERATING PROCEDURES

Procedures for pilots-in-command at the scene of an accident

- a) Keep the craft in distress in sight until compelled to leave the scene or advised by the rescue coordination centre that it is no longer necessary.
- b) Determine the position of the craft in distress.
- c) Report to the rescue coordination centre or air traffic services unit as much information as possible.

SATELLITE AIDED SEARCH AND RESCUE

India has evolved a Satellite-aided Search and Rescue programme participation in the COSPAS/SARSAT systems. It operates on 406MHz. Location accuracy is normally within 5km. The system will detect transmissions on this frequency throughout the Indian Search and Rescue Region (SRR) and also SSR of Bangladesh, Myanmar, Bhutan, Indonesia, Kenya, Malaysia, Maldives, Mauritius, Nepal, Seychelles, Singapore, Somalia, Sri Lanka, Thailand and Tanzania.

Under this programme two local user terminals (LUT) have been established at Bangalore and Lucknow with the Indian Mission Control Centre (INMCC) at Bangalore which would be responsible for coordination with Rescue coordination Centre and other International Mission Control Centre.

The INMCC at Bangalore is connected with the RCC's at Mumbai, Delhi, Kolkata and Chennai through AFS network.

RUNWAY/TAXIWAY LIGHTS AND MARKINGS

VISUAL AIDS FOR NAVIGATION

INDICATORS AND SIGNALLING DEVICES WIND DIRECTION INDICATORS (WDI)

Application

An aerodrome shall be equipped with at least one wind direction indicator.

Location

A wind direction indicator shall be located so as to be visible from aircraft in flights on the movement area.

SIGNALLING LAMP

Application

A signalling lamp shall be provided at a controlled aerodrome in the aerodrome control tower.

Characteristics

Recommendation: A signalling lamp should be capable of producing red, green and white signals.

Runway markings shall be white.

Taxiway markings, runway turn pad markings and aircraft stand markings shall be yellow.

Apron safety lines shall be of red colour in India.

RUNWAY DESIGNATION MARKING

A runway designation marking shall be provided at the thresholds of a paved runway.

Characteristics

A runway designation marking shall consist of a two-digit number and on parallel runways shall be supplemented with a letter.

RUNWAY CENTRE LINE MARKING

A runway centre line marking shall be provided on a paved runway.

Location

A runway centre line marking shall be located along the centre line of the runway between the runway designation markings.

Characteristics

The length of a stripe plus a gap shall be not less than 50 m or more than 75 m. The length of each strips shall be at least equal to the length of the gap or 30 m, whichever is greater.

The width of the stripes shall be not less than:

- 0.90 m on precision approach category II and III runways.
- 0.45 m on non-precision approach runways where the code number is 3 or 4 and precision approach category I runways.
- 0.30 m on non-precision approach runways where the code number is 1 or 2, and on non-instrument runways.

Location

The stripes of the threshold marking shall commence 6 m from the threshold.

Characteristics

The number of strips shall be in accordance with the runway width as follows; -

Runway Width	Number of Stripes
18m	4
23m	6
30m	8
45m	12
60m	16

ARROWS

Where a runway threshold is permanently displaced, arrows conforming shall be provided on the portion of the runway before the displaced threshold.

Application

An aiming point marking shall be provided at each approach end of a paved instrument runway where the code number is 2, 3 or 4.

Location and dimensions

Landing distance available

Less than 800m Including 1200m	800m up to but not including 2400m	1200m up to but not including 2400m	2400m and above	
Distance from threshold	150m	250m	300m	400m
To beginning of marking				
Length of strips 60m	30-40m	30-45m	45-60m	45-
Width of strips 10m	4m	6m	6-10m	6-
Lateral spacing between				
Inner sides of stripes 22.5m	6m	9m	18-22.5m	18-

Application

A touchdown zone marking shall be provided in the touchdown zone of a paved precision approach runway where the code number is 2,3 or 4.

Landing distance available or the distance	Pair(s) of marking
Between thresholds	
Less than 900m	1
900m up to but not including 1200m	2
1200m up to but not including 1500m	3
1500m up to but not including 2400m	4
2400m or more	6

TAXIWAY CENTRE LINE MARKING

Application

Taxiway centre line marking shall be provided on a paved taxiway, de-icing/anti-icing facility and apron where the code number is 3 or 4 in such a way as to provide continuous guidance between the runway centre line and aircraft stands.

Characteristics

A taxiway centre line marking shall be at least 15 cm in width and continuous in length except where it intersects with a runway-holding position marking or an intermediate holding position marking.

ISOLATED AIRCRAFT PARKING POSITION

An isolated aircraft parking position shall be designated or the aerodrome control tower shall be advised of an areas suitable for the parking of an aircraft which is known or believed to be the subject of unlawful interference.

RUNWAY-HOLDING POSITION MARKING

Application and location

A runway-holding position marking shall be displayed along a runway-holding position

Characteristics

A VOR aerodrome checkpoint marking shall consist of a circle 6m in diameter and have a line width of 15 cm.

Application

Aircraft stand markings should be provided for designated parking positions on a paved apron and on a de-icing/anti-icing facility.

Location

Apron safety lines shall be located so as to define the areas intended for use by ground vehicles and other aircraft servicing equipment etc. To provide safe separation from aircraft.

MANDATORY INSTRUCTION MARKING

Application

Where it is impracticable to install a mandatory instruction sign a mandatory instruction marking shall be provided on the surface of the pavement.

Location

The mandatory instruction marking shall be located on the left-hand side of the taxiway centre line marking and on the holding side of the runway-holding position marking.

Characteristics

A mandatory instruction marking shall consist of an inscription in white on a red background. Except for a NO ENTRY marking.

AERONAUTICAL BEACONS

Aerodrome beacon shall be provided at an aerodrome intended for use at night if one or more of the following conditions exists:

- a) Aircraft navigate predominantly by visual means.
- b) Reduced visibilities are frequent
- c) It is difficult to locate the aerodrome from the air due to surrounding lights or terrain.

Location

The aerodrome beacon shall be located on or adjacent to the aerodrome in an area of low ambient background lighting.

Characteristics

The aerodrome beacon shall show either coloured flashes alternating with white flashes, or white flashes only. Where used, the coloured flashes emitted by beacons at water aerodromes shall be yellow.

SIMPLE APPROACH LIGHTING SYSTEM

Location

A simple approach lighting system shall consist of a row of lights on the extended centre line of the runway extending, whenever possible, over a distance of not less than 420 m from the threshold with a row of lights forming a crossbar 18m or 30 m in length at a distance of 300m from the threshold.

The lights forming the centre line shall be placed at longitudinal intervals of 60m. Except that when it is desired to improve the guidance, an interval of 30 m may be used.

Characteristics

The lights of a simple approach lighting system shall be fixed lights.

A barrette at least 3 m in length

PRECISION APPROACH CATEGORY I LIGHTING SYSTEM

Location

A precision approach category I lighting system shall consist of a row of lights over a distance of 900m from the runway threshold, with a row of lights forming a crossbar 30 m in length at a distance of 300 m from the runway threshold.

The lights forming the centre line shall be placed at longitudinal intervals of 30 m with the innermost light located 30 m from the threshold.

PRECISION APPROACH CATEGORY II AND III LIGHTING SYSTEM

Location

The approach lighting system shall consist of a row of lights on the extended centre line of the runway, extending, wherever possible over a distance of 900 m from the runway threshold. In addition, the system shall have two side rows of lights, extending 270m from the threshold, and 2 crossbars, 1 at 150 m and 1 at 300 m from threshold.

The lights forming the centre line shall be placed at longitudinal intervals of 30m, with the innermost lights located 30m from the threshold.

The crossbar provided at 300m from the threshold shall extend on both sides of the centre line lights to a distance of 15m from the centre line.

T-VASIS AND AT-VASIS

Description

The T-VASIS shall consist of 20 light units symmetrically disposed about the runway centre line in the form of two wing bars and four light units each, with bisecting longitudinal lines of six lights.

The AT-VASIS shall consist of ten light units arranged on one side of the runway in the form of a single wing bar of four light units with a bisecting longitudinal line of six lights.

PAPI and APAPI

Description

The PAPI system shall consist of a wing bar of four sharp transition multi-lamp (or paired single lamp) units equally spaced. The system shall be located on the left side of the runway unless it is physically impracticable to do so.

The APAPI system shall consist of a wing bar of two sharp transition multi-lamp (or paired single lamp) units. The system shall be located on the left side of the runway unless it is physically impracticable to do so.

RUNWAY LIGHTING

RUNWAY THRESHOLD IDENTIFICATION LIGHTS

Characteristics

Runway threshold identification lights should be flashing white lights with a flash frequency between 60 and 120 per minute.

The lights shall be visible only in the direction of approach runway.

RUNWAY EDGE LIGHTS

Application

Runway edge lights shall be provided for a runway intended for use at night or for a precision approach runway intended for use by day or night.

Location

Runway edge lights shall be placed along the full length of the runway and shall be in two parallel rows equidistant from the centre line.

Runway edge lights shall be placed along the edges of the area declared for use as the runway or outside the edge, of the area at a distance of not more than 3m.

The lights shall be uniformly spaced in rows at intervals of not more than 60m for an instrument runway and at intervals of not more than 100m for a non-instrument runway.

Characteristics

Runway edge lights shall be fixed lights showing variable white except that in case of a displaced threshold the lights between the beginning of the runway and the displaced threshold shall show red in the approach direction.

RUNWAY THRESHOLD AND WING BAR LIGHTS

Application of runway threshold lights

Runway threshold lights shall be provided for a runway equipped with runway edge lights except on a non-instrument or non-precision approach runway where the threshold is displaced and wing bar lights are provided.

Location of runway threshold lights

When a threshold is at the extremity of a runway the threshold lights shall be placed in a row at right angles to the runway axis as near to the extremity of the runway as possible and in any case not more than 3 m outside the extremity.

When a threshold is displaced.

Threshold lighting shall consist of:

- a) On a non-instrument or non-precision approach runway at least six lights.
- b) On a precision approach runway category, I at least the number of the lights that would be required if the lights were uniformly spaced at intervals of 3m between the rows of runway edge lights.

Application of wing bar lights

Wing bar lights shall be provided on a non-instrument or non-precision approach runway where the threshold is displaced and runway threshold lights are required but are not provided.

Location of wing bar lights

Wing bar lights shall be symmetrically disposed about the runway centre line at the threshold in two groups, i.e. wing bars. Each wing bar shall be formed by at least five lights extending at least 10 m outward from and at right angles to the line of the runway edge lights.

Characteristics of runway threshold and wing bar lights

Runway threshold and wing bar lights shall be fixed unidirectional lights showing green in the direction of approach to the runway.

RUNWAY END LIGHTS

Application

Runway end lights shall be provided for a runway equipped with runway edge lights.

Location

Runway end lights shall be placed on a line at right angles to the runway axis as near to the end of the runway as possible and in any case, not more than 3 m outside the end.

Runway end lighting should consist of at least six lights.

The spacing between runway end lights except between the two innermost lights if a gap is used, should not exceed 6m.

Characteristics

Runway end lights shall be fixed unidirectional lights showing red in the direction of the runway.

RUNWAY CENTRE LINE LIGHTS

Application

Runway centre line lights shall be provided in a precision approach runway category II or III.

Runway centre line lights shall be provided on a runway intended to be used for take off with an operating minimum below an RVR of the order of 400m.

Characteristics

Runway centre line lights shall be fixed lights showing variable white from the threshold to the point 900 m from the runway end alternate red and variable white from 900 m to 300 m from the runway end and red from 300m to the runway end except that for runways less than 1800m in length the alternate red and variable white lights shall extend from the midpoint of the runway usable for landing to 300m from the runway end.

RUNWAY TOUCHDOWN ZONE LIGHTS

Application

Touchdown zone lights shall be provided in the touchdown zone of a precision approach runway category II or III.

Location

Touchdown zone lights shall extend from the threshold for a longitudinal distance of 900m except that on runways less than 1800m in length the system shall be shortened so that it does not extend beyond the midpoint of the runway. The lateral spacing between the innermost lights of a pair of barrettes shall be equal to the lateral spacing selected for the touchdown zone marking. The longitudinal spacing between pairs of barrettes shall be either 30m or 60m.

Characteristics

A barrette shall be composed of at least three lights with a spacing between the lights of not more than 1.5m

Touchdown zone lights shall be fixed unidirectional lights showing variable white.

SIMPLE TOUCHDOWN ZONE LIGHTS

The purpose of simple touchdown zone lights is to provide pilots with enhanced situational awareness. Lights shall be a pair of fixed unidirectional lights showing variable white, located on each side of the runway centreline, aligned so as to be visible to the pilot of a landing aeroplane in the direction of approach.

STOPWAY LIGHTS

Location

Stop way lights shall be placed along the full length of the stop way and shall be in two parallel rows that are equidistant from the centre line and coincident with the rows of the runway edge lights.

Characteristics

Stop way lights shall be fixed unidirectional lights showing red in the directions of the runway.

TAXIWAY LIGHTS

TAXIWAY CENTRE LINE LIGHTS

Application

Taxiway centre line lights shall be provided on an exit taxiway, taxiway, de-icing facility and apron intended for use in runway visual range conditions less than a value of 350m in such a manner as to provide continuous guidance between the runway centre line and aircraft stands.

Characteristics

Taxiway centre line lights on taxiway other than an exit taxiway and on a runway forming part of standard taxi-route shall be fixed lights showing green with beam dimensions such that the light is visible only from aeroplanes on or in the vicinity of the taxiway.

Taxiway centre line lights on an exit taxiway shall be fixed lights. Alternate taxiway centre line lights shall show green and yellow from their beginning near the runway centre line.

The light nearest to the perimeter shall always show yellow. Where aircraft may follow the same centre line in both directions, all the centre line lights shall show green to aircraft approaching the runway.

Location:

Taxiway centre line lights may be offset by not more than 30 cm where it is not practicable to locate them on the marking.

TAXIWAY CENTRE LINE LIGHTS ON TAXIWAYS

Location

Taxiway centre line lights on a straight section of a taxiway should be spaced at longitudinal intervals of not more than 30m

- On a taxiway intended for use in RVR conditions of less than a value of 350m, the longitudinal spacing should not exceed 15m.

TAXIWAY EDGE LIGHTS

Locational

Taxiway edge lights on a holding bay, de-icing/anti icing facility apron etc. Should be spaced at uniform longitudinal intervals of not more than 60m.

The lights should be located as near as practicable to the edges of the taxiway, holding bay, de-icing/anti-icing facility, apron or runway etc or outside the edges at a distance of not more than 3 m.

Characteristics

Taxiway edge lights shall be fixed lights showing blue. The lights shall show up to at least 30 degrees above the horizontal and at all angles in azimuth necessary to provide guidance to a point taxiing in either direction.

AIRCRAFT STAND MANOEUVRING GUIDANCE LIGHTS

Characteristics

Aircraft stand manoeuvring guidance lights other than those indicating a stop position, shall be fixed yellow lights, visible throughout the segments within which they are intended to provide guidance.

The lights indicating a stop position shall be fixed, unidirectional lights, showing red.

Runway Guard Lights

The purpose of runway guard lights is to warn pilots and drivers of vehicles when they are operating on taxiways, that they are about to enter runway.

Mandatory Instruction Signs

A mandatory instruction sign shall be provided to identify a location beyond which an aircraft taxiing or vehicle shall not proceed unless authorized by the aerodrome control tower.

Mandatory instruction signs shall include runway designation signs, category I, II or III holding position signs, runway-holding position signs, road-holding position signs and NO ENTRY signs.

A NO ENTRY sign shall be provided when entry into an area is prohibited.

A mandatory instruction sign shall consist of an inscription in white on a red background.

Information Signs

An information sign shall be provided where there is an operational need to identify by a sign a specific location or routing information.

Information signs shall include, direction signs, location signs, destination signs, runway exit signs, runway vacated signs and intersection take-off signs.

A runway exit sign shall be provided where there is an operational need to identify a runway exit.

An intersection take-off sign shall be provided when there is an operational need to indicate the remaining take-off run available (TORA) for intersection take-offs.

Location Sign

A location sign should be provided at an intermediate holding position.

A location sign shall be provided in conjunction with a runway designation sign except at a runway/runway intersection.

An information sign other than a location sign shall consist of an inscription in black on a yellow background.

A location sign shall consist of an inscription in yellow on a black background and where it is a stand-alone sign have a yellow border. Remember BLACK SQUARE, YOU ARE THERE.

VOR aerodrome check-point sign

A VOR aerodrome check-point sign shall consist of an inscription in black on a yellow background.

Aerodrome identification sign

Aircraft stand identification signs

Aircraft stand identification signs

An aircraft stand identification sign shall consist of an inscription in black on a yellow background.

Road holding position sign

A road-holding position sign shall be provided at all road entrances to a runway.

A road-holding position sign shall consist of an inscription in white on a red background.

Low-intensity obstacle light

Low-intensity obstacle lights, displayed on vehicles associated with emergency or security shall be flashing blue and those displayed on other vehicles shall be flashing yellow.

Low-intensity obstacle lights displayed on follow-me vehicles shall be flashing yellow.

Emergency and other Services: Rescue and Fire Fighting

The level of protection provided at an aerodrome for rescue and firefighting shall be appropriate to the aerodrome category determined using the principles given below, except that where the number of movements of the aeroplanes in the highest category normally using the aerodrome is less than 700 in the busiest consecutive three months, the level of protection provided shall be not less than one category below the determined category.

Either a take-off or a landing constitutes a movement.

Extinguishing agents-

The principal extinguishing agent should be foam.

The complementary extinguishing agent should be a dry chemical powder suitable for extinguishing hydrocarbon fires.

Aerodrome category for rescue and fire fighting

Aerodrome category Width	Aeroplane overall length	Maximum fuselage
(1) (2)	(3)	
1	0 m up to but not including 9 m	2 m
2	9 m up to but not including 12 m	2m
3	12 m up to but not including 18 m	3m
4	18 m up to but not including 24 m	4m
5	24m up to but not including 28m	4m
6	28m up to but not including 39m	5m
7	39m up to but not including 49m	5m
8	49m up to but not including 61m	7m
9	61m up to but not including 76m	7m
10	76m up to but not including 90m	8m

Response Time

A response time not exceeding three minutes shall be maintained by the rescue and firefighting services for any point of each operational runway in optimum visible and surface conditions.

The operational objective of the rescue and firefighting services shall be to achieve a response time not exceeding two minutes to any point of each operational runway, in optimum visibility and surface conditions.

Communication and Alerting systems

A discrete communication system should be provided linking a fire station with the control tower.

PROCEDURES FOR AIR NAVIGATION SERVICES

AIRCRAFT OPERATIONS

(DOC 8168, Vol. 1 – Flight Procedures)

INTRODUCTION

The Procedures for Air Navigation Services – Aircraft Operations (PANS-OPS) –Doc 8168 consists of two volumes as follows:

Volume I – Flight Procedures describes operational procedures recommended for the guidance of flight operations personnel and flight crew.

Volume II – Construction of Visual and Instrument Flight Procedures

GENERAL CRITERIA FOR DEPARTURE PROCEDURES

Procedures contained in PANS-OPS assume that all engines are operating.

Note – Development of contingency procedures is the responsibility of the operator.

INSTRUMENT DEPARTURE PROCEDURE

Aerodrome operating minima-

- Where obstacles cannot be cleared by the appropriate margin when the aeroplane is flown on instruments, aerodrome operating minima are established to permit visual flight clear of obstacles.
- When a departure route requires a turn of more than 15 degrees to avoid an obstacle, a turning departure is constructed.

OBSTACLE CLEARANCE

The minimum obstacle clearance equals zero at the departure end of the runway (DER). From that point, it increases by 0.8 per cent of the horizontal distance in the direction of flight assuming a maximum turn of 15 degrees.

In the turn initiation area and turn area a minimum obstacle clearance of 90 m (295 ft) is provided.

STANDARD INSTRUMENT DEPARTURES

SID Termination

The SID terminates at the first fix/facility/waypoint of the en-route phase following the departure procedures.

STRAIGHT DEPARTURES

A straight departure is one in which the initial departure track is within 15 degrees of the alignment of the runway centre line.

TURNING DEPARTURES

When a departure route requires a turn of more than 15 degrees, it is called a turning departure. Straight flight is assumed until reaching an altitude/height of at least 120m (394 ft). Procedures normally cater for turns at a point 600m from the beginning of the runway. However, in some cases turns may not be initiated before the DER and this information will be noted on the departure chart.

There are four kinds of turns:

- a) Turn at a fly-by waypoint
- b) Turn at a flyover waypoint
- c) Turn at an altitude/height
- d) Fixed radius turn (generally associated with procedures based on RNP)

TYPES OF APPROACH

Straight-in Approach

Whenever possible, a straight-in approach will be specified which is aligned with the runway centre line. In the case of non-precision approach, a straight-in approach is considered acceptable if the angle between the final approach track and the runway centre line is 30 degrees or less.

Circling Approach

A circling approach will be specified in those cases where terrain or other constraints cause the final approach track alignment or descent gradient to fall outside the criteria for a straight-in approach.

CATEGORIES OF Aircraft

Category A: less than 169 km/h (91 kt) indicated airspeed (IAS)

Category B: 169 km/h (91 kt) or more but less than 224 km/h (121 kt) IAS

Category C: 224 km/h (121 kt) or more but less than 261 km/h (141 kt) IAS

Category D: 261 km/h (141 kt) or more but less than 307 km/h (166 kt) IAS

Category E: 307 km/h (166 kt) or more but less than 391 km/h (211 kt) IAS

Segments of the Approach Procedure

The approach segments begin and end at designated fixes. However, under some circumstances certain of the segments may begin at specified points where no fixes are available. For eg. The final approach segment of a precision approach may start where the intermediate flight altitude intersects the nominal glide path (the final approach point).

DESCENT GRADIENT

In instrument approach procedure design, adequate space is allowed for descent from the facility crossing altitude/height to the runway threshold for straight-in approach or to OCA/H for circling approaches.

FAF is 5.2 per cent/3.0dg (52m/km 318ft/NM). Where a steeper descent gradient is necessary, the maximum permissible is 6.5 per cent/3.7dg (65m/km 395ft/NM) for category A and B aircraft, 6.1 per cent/3.5dg (61m/km 370ft/NM) for category CD and E aircraft and 10 per cent (5.7dg) for category H.

CONTINUOUS DESCENT FINAL APPROACH (CDFA)

The technique requires a continuous descent, flown either with VNAV guidance calculated by on-board equipment or based on manual calculation of the required rate, of descent, without level-offs. The rate of descent is selected and adjusted to achieve a continuous descent to a point approximately 15m (50 ft) above the landing runway threshold.

ARRIVAL SEGMENT

A standard instrument arrival (STAR) route permits transition from the en-route phase to the approach phase.

MINIMUM SECTOR ALTITUDES (MSA)/TERMINAL ARRIVAL ALTITUDES (TAA)

Minimum sector altitudes or terminal arrival altitudes are established for each aerodrome and provide at least 300 m (1000 ft) obstacle clearance within 46 km (25 NM) of the navigation aid.

TYPES OF MANOEUVRES

Where no suitable IAF or IF is available to construct the instrument procedure, a reversal procedure, racetrack or holding pattern is required.

Reversal Procedure

The reversal procedure may be in the form of a procedure or base turn. Entry is restricted to a specific direction or sector. In these cases, a specific pattern – normally a base turn or procedure turn – is prescribed.

The directions and timing specified should be strictly followed in order to remain within the airspace provided.

Racetrack Procedure

A racetrack procedure consists of:

- a) A turn from the inbound track through 180dg, from overhead the facility or fix on to the outbound track, for 1,2 or 3 minutes followed by
- b) A 180dg. turn in the same direction to return to the inbound track.

INTERMEDIATE APPROACH SEGMENT

Beginning and end of the Segment

Where a final approach fix (FAF) is available, the intermediate approach segment begins when the aircraft is on the inbound track of the procedure turn, base turn or final inbound leg of the racetrack procedure. It ends at the FAF or final approach point (FAP) as applicable.

Where no FAF is specified, the inbound track is the final approach segment.

FINAL APPROACH SEGMENT

Types of final approach

The criteria for final approach vary according to the type. These types are-

- a) Non-precision approach (NPA) with final approach fix (FAF)
- b) NPA without FAF
- c) Approach with vertical guidance (APV)
- d) Precision approach (PA)

NPA WITH FAF

FAF Location

This segment begins at a facility or fix, called the final approach fix (FAF) and ends at the missed approach point (MAPt). The optimum distance for locating the FAF relative to the threshold is 9.3 km (5.0 NM). The maximum length should not normally be greater than 19 km (10NM)

NPA WITHOUT FAF

Sometimes an aerodrome is served by a single facility located on or near the aerodrome, and no other facility is suitably situated to form a FAF. In this case, a procedure may be designed where the facility is both the IAF and the MAPt.

PRECISION APPROACH

Final approach point (FAP)

The final approach segment begins at the final approach point(FAP). This is a point in space on the final approach track where the intermediate approach altitude/height intercepts the nominal glide path.

Final approach Length

The intermediate approach altitude/height generally intercepts the glide path/MLS elevation angle at heights from 300m(1000ft) to 900m (3000ft) above runway elevation. In this case for a 3dg glide path, interception occurs between 6km (3NM) and 19 km (10NM) from threshold.

Outer Marker/DME fix

The final approach area contains a fix or facility that permits verification of the glide path. The outer marker or equivalent DME fix is normally used for this purpose.

MISSED APPROACH SEGMENT

CONSISTS OF THREE PHASES- initial, intermediate and final

It is designed to provide protection from obstacles throughout the missed approach manoeuvre. It specifies a point where the missed approach begins, and a point or an altitude/height where it ends.

The missed approach should be initiated not lower than the decision altitude/ height (DA/H) in precision approach procedures, or at a specified point in non-precision approach procedures not lower than the minimum descent altitude/height (MDA/H)

Missed Approach Gradient

Normally procedures are based on a minimum missed approach climb gradient of 2.5 per cent.

INITIAL PHASE

The initial phase begins at the MAPt and ends at the start of climb (SOC) no turns are specified in this phase.

INTERMEDIATE PHASE

The intermediate phase begins at the SOC.

FINAL PHASE

The final phase begins at the point where 50 m (164 ft) obstacle clearance is first obtained.

Instrumental approach operations shall be classified –

- a) Category I (CAT I) : a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800m runway visual range not less than 550m.
- b) Category II (CATII) : a decision height lower than 60m (200ft) but not lower than 30 m (100ft) and a runway visual range not less than 300m.
- c) Category IIIA (CAT IIIA) : a decision height lower than 30 m (100 ft) or no decision height and a runway visual range not less than 175 m .
- d) Category IIIB (CAT IIIB) : a decision height lower than 15 m (50ft) or no decision height and a runway visual range less than 175m but not less than 50 m .
- e) Category IIIC (CAT IIIC) : no decision height and no runway visual range limitations.

HOLDING PROCEDURES

IN – FLIGHT PROCEDURES ENTRY, HOLDING

Bank angle/rate of turn

All turns are to be made at a bank angle of 25dg or at a rate of 3dg per second, whichever requires the lesser bank.

Allowance for known wind

All procedures depict tracks. Pilots should attempt to maintain the track by making allowance for known wind by applying corrections both to heading and timing. This should be done during entry and while flying in the holding pattern.

ENTRY

The entry into the holding pattern shall be according to heading in relation to the three entry sectors. Zone of flexibility of 5dg. On either side of the sector boundaries.

Sector 1 entry – parallel entry

Sector 2 entry – offset entry

Sector 3 entry – direct entry

Transition altitude: A transition altitude shall normally be specified for each aerodrome by the State in which the aerodrome is lowest transition altitude in India is 4000 ft.

NATIONAL LAW

THE AIRCRAFT ACT. 1934

It extends to the whole of India and applies also:

- a) To citizens of India wherever they may be
- b) To and to the persons on, aircraft registered in India wherever they may be.
- c) To, and to the persons on aircraft, registered outside India but for the time being in or over India.
- d) To an aircraft operated by a person who is not a citizen of India but has his principal place of business or permanent residence of India.

Power of Central Government to exempt certain aircraft:

The Central Government may, by notification in the Official Gazette, exempt from all or any of the provisions of this Act any aircraft or class of aircraft and any person or class of persons, or may direct that such provisions shall apply to such aircraft or persons subject to such modifications as may be specified in the notification.

Power to detain aircraft:

- 1) Any authority authorised in this behalf by the Central Government may detain any aircraft, if in the opinion of such authority –
 - a) Having regard to the nature of an intended flight, the flight of such aircraft would involve danger to persons in the aircraft or to any other persons or property.
 - b) Such detention is necessary to secure compliance with any of the provisions of this Act.

Penalty for act in contravention of rule made under this Act: -

- 1) Prohibiting or regulating the carriage in aircraft of arms, explosives or other dangerous goods, or when required under the rules made under that clause to give information in relation to any such goods gives information which is false and which he either knows or believes to be false or does not believe to be true he, and if he is not the owner, the owner shall be punishable with imprisonment which may extend to two years and shall also be liable to fine which may extend to ten lakh rupees.
- 1A) Prohibiting the slaughter and flaying of animals and of depositing rubbish, filth and other polluted and obnoxious matter within a radius of 10 km from the aerodrome reference point, he shall be punishable with imprisonment which may extend to 3 yrs. Or with fine which may extend to 10 lakh rupees. Or with both.

11B. Penalty for failure to comply with directions issued under section 9A: -

- a) If any person fails to demolish any building, or structure or cut any tree or fails to reduce the height of any building, structure or tree in pursuance of any direction contained in any notification issued under sub-section (1) of section 9A within the period specified in the notification then subject to such rules as it shall be competent for any officer authorised by the Central Government in this behalf to demolish such building or structure or cut such tree or reduce the height of such building, structure or tree.

Areas over which flight by aircraft is prohibited:

- 1) The area included within a radius of one mile from the Towers of Silence on Malabar Hills.
- 2) The area near Baroda aerodrome extending vertically from ground level to an unlimited upper level.
- 3) The area around Rashtrapati Bhavan extending vertically from ground level to an unlimited upper level.
- 4) The area included within a radius of 10km from Mathura Refineries extending vertically from ground level to an unlimited upper level.
- 5) The area near Bhubaneswar extending vertically from ground level to an unlimited upper level of 50,000ft.

- 6) The area included within a radius of 10km from kalpakkam nuclear extending vertically from ground level up to an upper level of 10,000ft.

21. Dangerous flying: -

No person shall fly any aircraft in such circumstances as by reason of low altitude or proximity to persons or dwellings or for other reason, to cause unnecessary danger to any person or property.

22. Assault and other acts of interference against a crew member: - no person shall on board an aircraft-

Assault, intimidate or threaten, whether physically or verbally, a crew member.

24. Prohibition on consumption of intoxicating and psychoactive substances: -

No person acting as or carried in aircraft for the purpose of acting as pilot, commander, navigator, engineer, cabin crew or other operating member of the crew thereof, shall have taken or used any alcoholic drink, sedative, narcotic or stimulant drug or preparation within 12hrs. Of the commencement of the flight.

24 A. Carriage of persons suffering from mental disorders or epilepsy aircraft: -

No person shall knowingly carry or permit to be carried, or connive at the carriage of, a person suffering from any mental disorder or epilepsy.

Provided that this prohibition shall not apply if the person to be carried is certified by a registered medical practitioner to be fit to travel by air without being a risk to other passengers or to the aircraft.

24 B. Carriage of prisoners in aircraft: -

No prisoner shall be taken abroad or carried on an aircraft except under and in accordance with a permit in writing issued by the Director-General, a Deputy Director General, the Director of Regulations and Information or any other officer of the Civil Aviation Department authorised.

24 C. Carriage of animals, birds and reptiles in aircraft: -

No animal, bird or reptile shall be taken abroad or carried on any aircraft to, from and within India, except under and in accordance with a general or special permit in writing issued by the Director-General in this behalf.

25. Smoking in aircraft: -

The owner or the operator and the pilot-in-command of every aircraft registered in India, shall exhibit or cause to be exhibited in prominent place(s) in the aircraft notice(s) stating where and to what extent smoking is prohibited or permitted therein.

25 A. Fuelling of aircraft: -

The following precautions shall be observed-

- a) Fuelling of aircraft shall be done outdoors and not less than 15 meters from any building.
 - b) A "NO SMOKING" notice shall be prominently displayed.
 - c) Smoking or use of an appliance employing naked flame or use of an appliance capable of producing a spark or in any other way igniting fuel vapours shall not be permitted within 30 meters of the aircraft or fuelling equipment.
 - d) Aircraft engines shall not be started or turned and ignition switches shall be placed in the OFF position.
- "Danger Zone" is defined as the area within the largest polygon obtained by joining points 3 meters away from the wings and the fuelling vehicle.

- No person other than the staff of the operator, fuelling company and officials of the Civil Aviation Department, Customs and Police, shall be permitted within 15 meters of the aircraft.
- Passengers may be permitted to embark, disembark or remain in the cabin subject to the following conditions-
 - 1) An attendant shall remain on duty in the cabin of the aircraft. The attendant shall ensure that no smoking takes place or other source of ignition is allowed to occur.
 - 2) Fuelling operations shall cease when a turbo-jet aircraft manoeuvres so as to bring the rear jet outlets within 43 meters of the fuelling equipment or the aircraft.

25 B. Housing of aircraft: -

No aircraft containing dangerous petroleum in bulk in any of its tanks may be housed in a hangar unless such hangar is constructed of unflammable material and safely ventilated.

26. Dropping of articles and descent by parachutes: -

- No person shall drop or project or cause or permit to be dropped or projected from an aircraft in motion anything except ballast in the form of fine sand or water.
- The dropping of separate sheets of paper containing printed matter or separate petals of flowers in any place if the prior written permission of the District Magistrate or the commissioner of Police is obtained in each case.

27. Carriage of persons in unauthorised parts of aircraft: -

- No person shall at any time be carried on the wings or undercarriage of the aircraft.

28. Minimum age for sole control of aircraft

- No person being under 16yrs shall have sole control of an aircraft in motion.

28A. Maximum age limit for professional pilots: -

- No person holding a pilot's license issued under these rules and having attained the age of 65yrs. shall act as pilot-in-command or co-pilot of an aircraft engaged in commercial air transport operations.

29A. Prohibition of operating civil aircraft causing sonic boom: -

- No person shall operate a civil aircraft at a true flight Mach number greater than one over the territory of India or over the high seas in a manner which may cause or is likely to cause sonic boom over the territory of India.

29B. Prohibition on the use of portable electronic devices: -

No person shall operate, nor shall the operator or the pilot-in-command of an aircraft allow the operation of any portable electronic devices on board an aircraft in flight.

Provided that the Pilot-in-command may permit the use of cellular telephone by the passengers of a flight after the aircraft has landed and cleared active runway, except when the landing takes place in low visibility conditions as may be determined by the Director-General from time to time.

29D. Safety Management Systems: -

- Establish and maintain Safety Management Systems.
- Prepare a Safety Management Systems Manual in such form and manner as may be specified by the Director-General and submit the same to the Director-General for approval.

37A. Use of State Marks: -

- An aircraft shall not bear on any part of its exterior surface any advertisement or any sign or lettering except those under these rules and as required or permitted by the Director-General.
- National flags or colours may be displayed on the aircraft in such a manner that they are distinct and are not likely to create confusion with the markings used by military aircraft.

38. Licensing Authority: -

The authority by which the licences and ratings specified below may be granted, renewed or varied shall be the Central Government.

38A. Carriage of operating crew: -

PILOT'S

- a) Private Aircraft- Every private aircraft shall be flown by a person holding a valid Pilots License.
 - 1) A private aircraft shall not be flown by a person holding a Student Pilot's License.
 - 2) A private aircraft shall not be flown by a person holding a Private Pilot's License for remuneration or hire of any kind.
 - 3) A private aircraft carrying passengers at night shall not be flown by a person holding a Private Pilot's License, without having a valid Night Rating.
- b) Public transport - Every Public transport aircraft shall be flown by a person holding an appropriate professional pilots licence, commercial, senior commercial or Airline Transport Pilot's License issued in accordance with Schedule II.

FLIGHT INSTRUCTOR OR ASSISTANT FLIGHT INSTRUCTOR

- a) Every aircraft which is being used for the purpose of giving dual instructions in piloting shall carry a person holding an appropriate professional pilots license which has an appropriate Flight Instructors or Assistant Flight Instructors rating.
- b) No person other than a person having a Flight Instructors or Assistant Flight Instructors rating shall impart instructions in piloting an aircraft, unless he has been specifically authorised in writing by the Director General to impart such instructions.

FLIGHT NAVIGATOR

Every public transport aircraft engaged on a flight without landing over a great circle distance of more than 600 NMs and not equipped with the navigational equipment capable of providing instant and continuous ground position shall carry on board a Flight Navigator.

38B – carriage of Cabin Crew

42. Licences and their renewal –

- for a period exceeding 2 yrs. But not exceeding 3 yrs. Test of skill and Air Regulation.
- for a period exceeding 3 yrs. All examinations and tests required for the issue of the licence or rating.
- the holder of the licence shall not exercise the privileges of his licence without being declared fit after a fresh medical examination in the event of his having
- a sickness or injury involving incapacity for a period of 15 days or more for the work for which he is licensed.

42A- Pilot not to fly more than 125 hours during any period of 30 consecutive days: -

No pilot of a flying machine, shall in his capacity as such pilot, fly for more than 125 hours during any period of 30 consecutive days.

Explanation- For the purposes of this rule, the flying time of a pilot either as solo pilot or pilot-in-command of an aircraft will be counted fully and the flying time of a pilot engaged as co-pilot or supernumerary pilot will be counted at 80 percent of the flight time.

50.Certificate of airworthiness-

A certificate of airworthiness shall remain in force for such period as may be specified in the certificate and may from time to time be renewed by the Director-General.

52. Modifications and repairs-

A person shall not carry out any modification or repair affecting safety of any aircraft.

65. Aeronautical beacon and aeronautical ground lights: -

No person shall wilfully or negligently endanger or interfere with any aeronautical beacons or aeronautical ground light established or maintained by or with the approval of the Central Government or any light exhibited therefrom.

67A. Log Books of Flight Crew Personnel and logging of flight time: -

-all flight times shall be logged therein.

-all entries in log books shall be made in ink.

-log books shall not preserve for not less than 5yrs. after the date of the last entry therein.

-at the end of every quarter in a year, that is at the end of March, June, September and December, log books shall be certified for correctness of entries therein.

-by competent authorities such as the Operations Manager, Deputy Operations Manager or Chief Pilot of the Company.

-by the Chief Flying Instructor in the case of members of the club.

- the holder of a Private Pilot's License may log as pilot-in-command only the flight time during which he acts as pilot-in-command.

68. Notifications of Accidents: -

- an accident in which an aircraft is involved shall be notified in accordance with the provisions given below-

-the notice and information shall be sent as soon as possible and by the quickest means available and in any case within 24 hrs after occurrence of the accident.

70. Removal and preservation of damaged aircraft-

-the aircraft or any parts or contents thereof may be removed or interfered with so far as may be necessary for the purpose of extricating persons or animals dead or alive, of preventing the destruction of the aircraft and its contents by fire or other cause or of preventing any damage or obstruction to the public or to the air navigations or to other transport.

-passengers and crews personal luggage may be removed from the aircraft under the supervision of an Officer of Police Department, a Magistrate, an Officer of the Civil Aviation Department.

140B. Operations Manual: -

- An operation manual in the form approved by the Director-General shall be maintained by the Corporation.
- A copy of the operations manual or such part of the manual as may be prescribed by the Director-General shall be carried in all aircraft engaged in scheduled air transport services.

140C. Route Guide

-all aircraft of the Corporation engaged in scheduled air transport services shall carry a Route Guide, which shall in addition to any relevant information.

THE INDIAN AIRCRAFT RULES, 1920

Customs aerodrome and customs officers: -

-by notification in the Official Gazette declare any aerodrome in India to be a customs aerodrome.

Examination: -

-the person in-charge or any aircraft shall permit any customs collector or other officer authorised in this behalf by the Customs Collector at any time to board and examine the aircraft and any goods laden thereon.

AIRCRAFT MANUAL (INDIA) VOL I

-The Aircraft Act.1934 – An act for the control of manufacture, possession, use, operation, sale, import and export of aircraft.

-The Aircraft Act.1937 – Rules made under the act for regulating Civil Aviation activity in India. For details see Vol II Air Regulations.

The Indian Aircraft Rules.1920- These are custom rules governing aircraft arriving in or departing from India

THE AIRCRAFT (PUBLIC HEALTH RULES), 1954

Rules made to protect public health. The salient features of the rules are:-

- The commander of an aircraft on way to India shall inform the health officer at destination at least 2 hrs in advance on R/T detail of any person suffering from any sickness, quarantinable or infectious disease.
- Incubation period for infectious diseases is:
 - 5 Days – Influenza pneumonia, Cholera
 - 6 Days – Yellow fever, plague
 - 7 Days - Diphtheria
 - 8 Days – Relapsing fever
 - 10 Days- Cerebrospinal Meningitis
 - 14 Days- Small pox, Chicken Pox, Typhus

Dead bodies/human remains of persons who have died with diseases which cannot be brought into India- Yellow fever, Plague, Anthrax, Landers

(However, ashes of properly cremated bodies/remains can be brought)

Dead bodies/remains of persons who have died of any other reason can be brought provided they are properly enclosed in suitable coffin and other procedures are followed.

The Aircraft (Demolition of Obstructions caused by Building and Trees etc.) Rules 1994

Procedure for demolition of obstructions around aerodromes are outlined in these rules

The Aircraft (Carriage of Dangerous Goods) Rules 2003

Procedures for carriage of dangerous goods are outlined in these rules.

Aircraft Security Rules 2011

Deals with airport security

Aircraft (Investigation of Accidents and Incidents) Rules 2012

Deals with accident and incidents investigation

PERSONNEL LICENSING (ANNEX 1, AIRCRAFT RULES, 1937)

NATIONAL PROVISIONS

Licensing authority:

The authority by which the licences and ratings may be granted, renewed or varied shall be the Central Government which may withhold the grant or renewal of a licence or a rating, if for any reason it considers it desirable to do so.

Educational Qualification:

A) The minimum educational qualification for the following licences shall be a pass in Class Ten from a recognised Board. Student pilot's Licence, private pilot's licence, flight radio telephone operator's licence.

B) The minimum educational qualification for the following licences shall be a pass in class Ten plus Two, with Physics and Mathematics from a recognised Board/university or its equivalent examination.

Commercial pilot's licence

Airline Transport Pilot's Licence

Cross-country flights

- a) A cross-country flight unless otherwise stated, means a flight to a point beyond a radius of one hundred nautical miles from the aerodrome of departure whether over land or sea.
- b) A cross-country flight, for the purpose of PPL shall mean a flight to a point beyond a radius of fifty nautical miles from the aerodrome of departure.

Language Proficiency: -

An application for the PPL, CPL, ATPL shall have the ability to speak and understand the English language used for radiotelephony communications to the level of proficiency specified by the Director General.

Medical assessment is valid for 24 months for PPL. It is valid for 12 months for ATPL, CPL. Wherever 12 months validity is specified it is reduced to six months after the age of 40yrs.

Validity of Licences:

Licences are valid from the date of issue. Renewal will be done from the date of expiry provided the application is made within one month of date of expiry.

PPL is valid for 10yrs. All other licences are valid for 5yrs.

The holder of a pilot's licence should inform the Authority of any illness which they are suffering which involves incapacity to understand those functions to which the licence relates throughout a period of 15 days or more.

Minimum Age:

Minimum age for SPL is 16yrs, PPL 17yrs, CPL 18yrs and ATPL 21yrs

STUDENT PILOT LICENCE

(AEROPLANES/HELICOPTERS/GLIDERS)

Knowledge:

He shall pass oral examination in Air Regulations, Air Navigation, Aviation Meteorology and Aircraft and Engines as per the syllabus prescribed by the Director-General.

Security Clearance:

Security clearance for the Student Pilot/Pilot will be obtained from the concerned State Police authorities by the Flying Club/Government Flying Training School before the commencement of flying training and furnished to the Director-General.

Renewal:

The licence may be renewed for a period not exceeding 12 months from the date of fresh medical examination subject to the total validity of the licence not exceeding 24 months from the date of issue.

Privileges:

The privileges of the holder of a Student Pilot's licence –

- a) He shall fly at all times under the authority and supervision of a Flight Instructor or an Approved Examiner.
- b) He shall fly under Visual Flight Rules only.
- c) He shall not carry passengers, animals and goods or fly for hire, reward or remuneration of any kind.
- d) He shall not undertake cross-country flights unless he has a minimum of 10 hours of solo flight time and has passed the examinations in Air Navigation and Aviation Meteorology.

The Student Pilot's licence shall be issued by a Flying Club/Government Flying Training School specifically authorised in this regard and Service.

PRIVATE PILOT'S LICENCE (AEROPLANE)

Knowledge-

He shall pass a written examination in Air Regulations, Air Navigation, Aviation Meteorology and Aircraft and Engines as per the syllabus

prescribed by the by the Director-General.

Experience-

He shall produce evidence of having satisfactorily completed as a pilot of an aeroplane not less than 40 hours of flight time which shall include:

- 1) Not less than 20 hours of solo flight time
- 2) Not less than 5 hours of cross-country flight time.
- 3) Not less than 10 hours of solo flight time completed within a period of 12 months.

Rating: -

Night rating-

- 1) He must have completed not less than 50 hours of flight time as Pilot in command and as sole manipulator of the controls including not less than 5 hours by night, which include a minimum of 5 take-offs and 5 landings carried out within the preceding 6 months of the date of application.
- 2) He must have completed a dual cross-country flight by night of at least one hundred nautical miles before he can be permitted to undertake sole cross country flights by night.
- 3) He must have completed not less than 5 hours of dual instructions in instrument flying which may include not more than two and a half hours on an approved synthetic flight trainer.

COMMERCIAL PILOT'S LICENCE (AEROPLANES)

Knowledge-

He shall pass a written examination in Air Regulation, Air Navigation, Meteorology and aircraft and engines.

Experience-

He shall produce evidence of having satisfactorily completed as a pilot of an aeroplane within a period of 5yrs immediately preceding the date of application for licence not less than 200 hours of flight time, which shall include-

- 1) Not less than 100 hours of flight time as Pilot-in-command of which not less than 15 hours shall have been completed within a period of 6 months immediately preceding the date of application for licence.
- 2) Not less than 20 hours of cross-country flight time as Pilot-in-command including a cross-country flight of not less than 3 hundred nautical miles in the course of which full stop landings at two different aerodromes shall be made.
- 3) Not less than 10 hours of instrument time of which not more than 5 hours may be on an approved simulator
- 4) Not less than 5 hours of flight time by night including a minimum of 10 take-offs and 10 landings as Pilot-in-command as carried out within 6 months immediately preceding the date of application for licence.

Other requirements-

He shall be in possession of a current Flight Radio Telephone Operator's Licence for operation of radio telephone apparatus on board an aircraft issued by the Director-General.

AIRLINE TRANSPORT PILOT'S LICENCE (AEROPLANE)

Knowledge-

HE shall pass a written and oral examination in Air Regulations, Air Navigation, Avionics (Radio Aids and Instruments), Aviation Meteorology.

Note- Holder of a current Commercial Pilot's Licence, shall not be required to pass the examinations in a Air Regulation.

Experience-

He shall produce evidence of having satisfactorily completed as a Pilot of an aeroplane not less than 1500 hours of flight time of which not less than 150 hours' time shall be in the preceding twelve months and his total flying experience shall include-

- 1) Not less than 500 hours of flight time as a Pilot-in-command or as Co-pilot performing under the supervision of a pilot who fulfils the flying experience requirements of a check pilot, the duties and functions of a pilot-in-command provided that at least 200 hours out of these shall be cross-country flight time, including not less than 50 hours of flight time by night.
- 2) Not less than 1000 hours of total cross-country flight time
- 3) Not less than 100 hours of flight time by night
- 4) Not less than 100 hours of instrument time under actual or simulated instrument conditions of which not less than 50 hours shall be in actual flight.
- 5) Not less than 10 hours of flight time completed within a period of six months immediately preceding the date of application for the licence.

AIRWORTHINESS OF AIRCRAFT and DOCS TO BE CARRIED

(Annex 8. Rules of the Air, 1937)

Type Certificate- The State of Design, upon receipt of satisfactory evidence that the aircraft type is in compliance with the design aspects of the appropriate airworthiness requirements, shall issue a Type Certificate to define the type design and to signify its approval.

NATIONAL PROVISIONS

Certificate of Airworthiness –

- 1) The owner or operator of an aircraft may apply to the Director-General for the issue of a certificate of airworthiness in respect of the aircraft or for the validation of a certificate of airworthiness issued elsewhere in respect of the aircraft.
- 2) The Director-General may validate a certificate of airworthiness in respect of any aircraft that may be imported.

Provided that-

- a) The airworthiness authority of the country in which the aircraft is manufactured has issued a certificate of airworthiness or such equivalent document.
- b) Subject to these rules, a certificate of airworthiness shall remain in force subject to such conditions as may be specified by the Director-General. In addition, the Director-General may require the aircraft to be inspected by a person authorised in this behalf by the Director-General or tested in flight, or to be so inspected and so tested and the owner or operator of the aircraft shall give all necessary facilities for such inspection and tests.

Cancellation, suspension or endorsement on Type Certificate –

If any time the Director-General is satisfied that there is a reasonable doubt to indicate that the safety of the aircraft is imperilled because of a defect in the aeronautical product of the aircraft, he may cancel suspend or endorse the Type Certificate issued.

17. OPERATIONAL PROCEDURE

Admission to Flight Deck

An operator must ensure that no person, other than a flight crew member assigned to a flight, is admitted to, or carried in the flight deck unless that person is :

- An operating crew member.
- A representative of the Authority responsible for certification, licensing or inspection if this is required for the performance of his official duties
- Permitted by and carried in accordance with instructions contained in the Operations Manual.

The final decision regarding the admission to the flight deck shall be the responsibility of the commander.

Portable electronic devices- An operator shall not permit any person to use and take all reasonable measures to ensure that no person does use, on board an aeroplane, a portable electronic device that can adversely affect the performance of the aeroplane's systems and equipment.

Documents to be carried –

No person in charge of any aircraft shall allow such aircraft to be flown unless the following valid documents, as applicable (in original or attested copies) are carried on board the aircraft-

- Certificate of Registration
- Certificate of Airworthiness
- Airworthiness Review certificate (ARC)
- A document attesting Noise Certificate of the aeroplane/helicopter
- Air Operator's Permit
- Appropriate Licences for each member of the flight crew.
- Aero mobile Radio operation Licence for Radio Communication apparatus.
- Journey Log Book or equivalent documents approved by the DGCA.
- Operations Manual
- Minimum Equipment List
- Flight Manual
- Cabin Crew Manual
- Cockpit and Emergency Check List
- Maintenance Release/Certificate to release to services
- LOPA (layout of Passenger arrangement)
- Emergency and Safety Equipment Layout
- Route guides
- Current and suitable navigation charts.
- Weight Schedule
- Load and Trim Sheet
- If carrying passengers, a list of their names
- If carrying cargo, a manifest and detailed declaration of the cargo.
- If carrying dangerous goods, a list of such goods. This list must be specifically brought to the notice of pilot-in-command before the flight.

Leasing

Dry lease - is when the aeroplane is operated under the AOC of the lessee.

Wet lease – is when the aeroplane is operated under the AOC of the lessor.

OPERATOR CERTIFICATE

Air operator Permit/certificate (AOC) –

A Permit/certificate authorising an operator to carry out specified commercial air transport operations.

State of the Operator-

The state in which the operator's principal place of business is located or, if there is no such place of business, the operator's permanent residence.

Validity of an AOC

The validity of an AOC shall depend upon the operator maintaining the original certification standards under the supervision of the state of the operator.

Carriage of Person with Reduced Mobility

No airline shall refuse to carry persons with disability or persons with reduced mobility and their assistive aids/devices, escorts and guide dogs, including their presence in the cabin.

Smoking on Board

The commander shall ensure that no person on board is allowed to smoke.

RVSM

For flights in defined portions of airspace where, based on Regional Air Navigation Agreement a reduced vertical separation minimum (RVSM) of 300m (1000 ft.) applied between FL 290 and FL 410 inclusive and aeroplane.

- Indicating to the flight crew the flight level being flown
- Automatically maintaining a selected flight level
- Providing an alert to the flight crew when a deviation occurs from the selected flight level. The threshold for the alert shall not exceed = 90 m (300 ft)
- Automatically reporting pressure altitude.

ALL-WEATHER OPERATIONS (AWO)

Low visibility Procedures (LVP) – Procedures applied at an aerodrome for the purpose of ensuring safe operations during Category II and III approaches and Low visibility take-offs.

Low visibility take-off (LVTO) – A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m.

All aeroplanes of a maximum certificated take-off mass of over 15000 kg which are required to be equipped with both a CVR and an FDR, shall be equipped with 2 combination recorders (FDR/CVR). One recorder shall be located as close to the cockpit as practicable and the other recorder located as far as practicable.

All FDRs shall be capable of retaining the information recorded during at least the last 25 hours of their operation CVRs and Data link recorders shall be capable of retaining the information recorded during at least the last two hours of their operation.

GPWS:

All turbo-jet aeroplanes of a maximum certificated take-off mass in excess of 5700 kg or authorised to carry more than nine passengers should be equipped with a forward-looking wind shear warning system and a ground proximity warning system which has a forward-looking terrain avoidance function.

ACAS II : All turbine-engine aeroplanes of a maximum certificated take-off mass in excess of 5700 kg or authorised to carry more than 19 passengers shall be equipped with an airborne collision avoidance system (ACAS II).

First Aid Kits

One or more first-aid kits as per the following table is carried for the use of cabin crew in managing incidents of ill health.

Passenger	First Aid Kits
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0-100	1
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101-199	2
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200-299	3
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300 or more	4
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Oxygen Supply

A flight to be operated at flight altitude at which the atmospheric pressure in personnel compartments will be less than 700hPa shall not be commenced unless sufficient stored breathing oxygen is carried to supply.

- All crew members and 10% passengers for any period in excess of 30 minutes that the pressure in the compartments occupied by them will be between 700 hPa and 620hPa
- The crew and passengers for any period that the atmospheric pressure in compartments occupied by them will be less than 620hPa.

If operated at flight altitudes at the atmospheric pressure is more than 376hPa and cannot descend safely within 4 minutes to a flight altitude at which the atmospheric pressure is equal to 620hPa, there shall be no less than a 10-minute supply for the occupants of the passenger compartment.

MINIMUM FUEL REQUIREMENTS

The amount of usable fuel to be carried shall as a minimum be based on the following data-

- a) Taxi fuel
- b) Trip fuel
- c) Contingency fuel- it shall be 5% of the planned trip fuel or of the fuel required from the point of in-flight re-planning, but in any case, shall not be lower than the amount required to fly for 5 minutes at holding speed at 450 m (1500 ft) above the destination aerodrome in standard conditions.
- d) Destination alternate fuel
 - Where 2 destination alternate aerodromes are required the amount of fuel which requires the greater amount of alternate fuel.
- e) Final reserve fuel
 - for a reciprocating engine aeroplane, the amount of fuel required to fly for 45 minutes.
 - for a turbine-engine aeroplane, the amount of fuel required to fly for 30 minutes at holding speed at 450 m (1500 ft) above aerodrome elevation in standard conditions.
- f) additional fuel

Hand Fire Extinguishers

Portable fire extinguishers provided shall be of a type which, when discharged, will not cause dangerous contamination of the air within the aeroplane. At least one shall be located

- The pilot's compartment
- Each passenger compartment that is separate from the pilot's compartment and that is not readily accessible to the flight crew.

Crash Axes and Crowbars

An operator shall not operate an aeroplane with a maximum certificated take-off mass exceeding 5700 kg or having a maximum approved passenger seating configuration of more than 9 seats unless it is equipped with at least one crash axe or crowbar located on the flight deck.

Megaphones

An operator shall not operate an aeroplane with a maximum approved passenger seating configuration of more than 60 and carrying one or more passengers unless it is equipped with portable battery-powered megaphones readily accessible for use by crew members during an emergency evacuation.

Automatic Emergency Locator Transmitter

All aeroplanes shall carry an automatic ELT

Life rafts and survival ELTs for extended over-water flights

Flight Time: The total time from the moment an aeroplane first moves for the purpose of taking off until the moment it finally comes to rest at the end of the flight.

Neighbouring Countries: Countries whose standard times falls within a band of 3 hours (1:30 hours on either side of India.)

REQUIREMENTS FOR DOMESTIC AND NEIGHBOURING COUNTRIES

Daily maximum flight time limitations during any 24 consecutive hours:

Crew Complement

Maximum Flight Time Limitations/

Max no. Of landings*

8 hours

Up to 6 landings

Two-Pilot Operation for Day Operations

9 hour /up to 3 landings

For Night Operations

9 Hours/up to 2 landings

*Maximum number of landings is further dependent on Flight Duty Period

CUMULATIVE FLIGHT TIME LIMITATIONS

Cumulative period	Flight Time Limitation (Hours)
In 7 consecutive days	35
In 30 consecutive days	125
In 365 consecutive days	1000

CONSECUTIVE NIGHT OPERATIONS

No operator operating passenger flight shall deploy a flight crew nor a flight crew shall undertake any duty between period embracing 0000 to 0500 hours local time if during the previous day he/she performed flight duty between the period embracing 0000 to 0500 hours local time.

RECORDS

To enable the operator to ascertain that the fatigue management system is functioning as intended and as approved, records shall be kept for 18 months of the duties performed and rest periods provided so as to facilitate inspection by the operator's authorised personnel and surveillance/audit by DGCA officers.

SPECIAL OPERATIONAL PROCEDURES AND HAZARDS

(GENERAL ASPECTS)

Minimum Equipment List

The operator shall include in the Operations Manual a minimum equipment list (MEL), approved by the DGCA which will enable the pilot-in-command to determine whether a flight may be commenced or continued from any intermediate stop should any instrument, equipment or system become inoperative.

Aeroplane flight Manual (AFM)

Aeroplanes shall be operated in accordance with the provisions of the Flight Manual approved by the State of design.

Fire/smoke

Combustion consists of three elements – oxygen, heat and fuel. The goal of firefighting is to eliminate at least one element from the fire, in order to extinguish it.

A fire will continue, unless:

- The fuel supply has been cut off
- There is no more oxygen available
- The temperature has been cooled below the flammability temperature.

Class A: Fires that involve wood, paper, cloth or plastic. Class A fires needed to be cooled. A water extinguisher or liquid containing a large percentage of water, for e.g. Coffee, tea, juice will extinguish a class A fires.

-Water/glycol extinguishers are the most effective for class A fires.

Class B: Fires that involve flammable liquid, hydraulic fluid, oil, tar or aircraft fuel.

This type of fire cannot be extinguished with water, foam, or halon fire extinguishers should be used to extinguish class B fires.

Class C- Fires that involve electrical equipment. This type of fire must be extinguished with a non-conducting mixture, in order to avoid electrocution and damage to electrical circuitry. Halon fire extinguishers are effective for class C fires.

Class D- Fires that involve flammable metals, such as sodium, magnesium, lithium and potassium. Special powder extinguishers are effective on class D fire, because of the possible chemical reaction between the burning and extinguishing agents.

- Never use Halon fire extinguishers on class D fires.
- Class A and C fires are the most commonly encountered fires onboard aircraft.

Fire Extinguishers:

- Halon/BCF (Green) – is the generic name for the group of bromochlorodifluoromethane (BCF) extinguishers that can be used for class A, B and C fires.
- Carbon dioxide (CO₂), (Black) for class B and C fires.
- Dry powder (DP) (Blue) for class D fires
- Dry chemicals (Blue) for class A, B and C fires
- Water solution (H₂O) (Red) for class A fires.

A halon extinguisher is 3 times effective as CO₂, extinguishers that contain the same amount of extinguishing agent.

WAKE TURBULENCE LONGITUDINAL SEPARATION MINIMA

ARRIVING AIRCRAFT

S.no.	Leading Aircraft	Following Aircraft	Separation Minima
1	HEAVY	MEDIUM	2 Minutes
2	HEAVY OR MEDIUM	LIGHT	3 Minutes

DEPARTING AIRCRAFT

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	HEAVY	LIGHT OR MEDIUM	2 Minutes
2	MEDIUM	LIGHT	2 Minutes

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	HEAVY	MEDIUM or LIGHT	3 Minutes
2	MEDIUM	LIGHT	3 Minutes

DISPLACED LANDING THRESHOLD

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1.	HEAVY Arrival	LIGHT or MEDIUM Departure	
2.	MEDIUM		
3.	HEAVY Departure	LIGHT or MEDIUM Arrival	
3.	MEDIUM Departure	LIGHT or MEDIUM Arrival	
4.	MEDIUM Departure	LIGHT Arrival	

Preceding Aircraft	Succeeding Aircraft	Distance Base Wake Turbulence
HEAVY	5 NM	
HEAVY	MEDIUM	5 NM
LIGHT	6NM	

DEALING WITH WAKE TURBULENCE

Wake Turbulence Characteristics: Wake turbulence is created only when an airplane develops lift. It is greatest when generating airplane is heavy, clean and slow. In wake turbulence the circulation of vortex is outward, upward and around each wing tip.

FUEL JETTISONING

An aircraft in an emergency or other urgent situation may need to dump fuel so as to reduce to maximum landing weight in order to affect a safe landing.

The level to be used, which should be not less than 1800 m (6000 ft.)

SEPARATION

1. At least 19 km (10NM) horizontally, but not behind the aircraft dumping fuel.
2. Vertical separation if behind the aircraft dumping fuel within 15 minutes flying time or a distance of 93 km (50NM) by-
 - At least 300m (1000 ft) if above the aircraft dumping fuel.
 - At least 900m (3000 ft) if below the aircraft dumping fuel.

DYNAMIC HYDROPLANING:

Aquaplaning also known as hydroplaning, is a condition that exists when landing on a surface with standing water deeper than the tread depth of the tires. When the brakes are applied, there is a possibility that the brake will lock up and the tire will ride on the surface of the water, much like a water ski. When the tires are hydroplaning, directional control and braking action are virtually impossible. An effective anti-skid system can minimize the effects of hydroplaning.

COMMUNICATIONS

VFR COMMUNICATIONS

Q-code groups commonly used in RTF air-ground communications:

QDL – I intend to ask you for a series of bearings

QDM- the MAGNETIC heading for you to steer to reach me (or...) with no wind was ...degrees (at...hour)

QFE- Your MAGNETIC bearing from me (or from...) was ... degrees (at...hours)

QFE-If you set the subscale of your altimeter to read ...millibars, the instrument would indicate its height above aerodrome elevation (above threshold, runway number).

QFU- The magnetic direction (or number) of the runway to be used is ...

Note- The runway number is indicated by a two-figure group and the magnetic direction by a three-figure group.

QGH- you may land using....(procedure or facility)

QNE-on landing at... (place) at.....hours, with your sub-scale being set to 1013.2 millibars (29.92 inches, your altimeter will indicate....(figures and units)

QNH-If you set the subscale of your altimeter to read millibars, the instrument would indicate its elevation if your aircraft were on the ground at my station at....hours.

Note- When the setting is given in hundredths of inch the abbreviation INS is used to identify the units.

QTE-Your TRUE bearing from me is ...degrees... at hours or

Your TRUE bearing from(call sign) was....degrees at ... hours or

The TRUE bearing of(call sign) from....(call sign) was....degrees at.....hours.

QUJ- The TRUE track to reach me (or...) is degrees at ... hours.

Call signs for Aeronautical Stations:

Unit/Service available	Call sign suffix
Area Control Centre	CONTROL
Approach control	APPROACH
Approach Control Radar Arrivals/Departure	ARRIVAL/DEPARTURE
Aerodrome Control	TOWER
Surface Movement Control	GROUND
Radar (in general)	RADAR
Precision Approach Radar	PRECISION
Direction-finding station	HOMER
Flight Information Service	INFORMATION
Clearance Delivery	DELIVERY

Apron Control	APRON
Company Dispatch	DISPATCH
Aeronautical Station	RADIO
Flow Control	FLOW

When the tests are made, the following readability scale should be used:

Scale	Readability
1	Unreadable
2	Readable now and then
3	Readable but with difficulty
4	Readable
5	Perfectly readable

CAV-OK : When the visibility is 10 km or more, there is no cloud below 1500 m (5000 ft) or below the highest minimum sector altitude, whichever is greater, and no cumulonimbus, and there is no precipitation, thunder storm, shallow fog or low drifting snow, the term “CAVOK” is used.

DISTRESS AND URGENCY PROCEDURES

Frequencies:

- 121.5/243 MHz – International VHF/UHF distress frequency
- 2182 KHz – International distress frequency (MF band)
- 500 KHz – International Maritime distress frequency
- Special purpose SSR Codes are 7700 (Emergency), 7600 (Radio Failure) and 7500 (Hijack or Other Act of Violence)
- ELT Transmissions are on 121.5 Mhz, 243 Mhz and 406 Mhz.

Band	Frequency Range	Wavelength Range	Wavelength Denomination
VLF-Very low Frequency	3-30 KHz	100km-10km	Myriametric
LF- Low Frequency	3-300 KHz	10km-1km	Kilometric
MF-Medium Frequency	300 KHz-3 MHz	1km-100km	Hectometric
HF-High Frequency	3-30 MHz	100m 10m	Decametric
VHF-Very High Frequency	30-30 MHz	10m-1m	Metric
UHF-Ultra High Frequency	300 MHz-3GHz	1m-10cm	Decimetric
SHF-Supe High Frequency	3-30GHz	10cm-1cm	Centimetric

(Microwave)

EHF-Extremely High Frequency	30-300GHz	1cm-1mm	Millimetric
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Ground Wave: A wave received either directly from transmitter or after reflection from ground is called a ground wave.

Sky Wave: A wave received at receiver after getting refracted from atmosphere(ionosphere)

System	Frequency	Frequency Band
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Decca	70 to 130 KHz	LF
Loran C	100 KHz	LF
ADF	190 to 1750 KHz	LF/MF
HF communications	2 to 25 MHz	HF
ILS Marker	75 MHz	VHF
ILS Localiser	108.1 to 111.95 MHz	VHF
VOR	108.0 to 117.95 MHz	VHF
VHF Communications	118 to 137 MHz	VHF
ILS Glidepath	329.15 to 335.0 MHz	UHF
DME	960 to 1213 MHz	UHF
SSR	1030 and 1090	UHF
GPS	1575.42 MHz (L1)	UHF
1227.6 MHz(L2)	UHF	
Satcome (Inmarsat)	1500 to 1600 MHz	UHF
(Aircraft to Satellite)		
4000 to 6000 MHz	SHF	
(Satellite to Ground)		
Radio Altimeter	4200 to 4400 MHz	SHF
Weather Radar	9375 MHz	SHF
MLS	5031 to 5091 MHz	SHF
ATC Surveillance	600 to 1300 MHz	UHF
Radars		
ATC Ground	10 to 16 GHz	SHF
Manoeuvre Radar		
LOS (range)=1.25(height + Ht of Rx)		

AIRCRAFT ACCIDENT AND INCIDENT

Aircraft Accident and Incident Investigation

- The accident investigation preliminary report shall be submitted to appropriate States and to the ICAO, in one of the working languages of ICAO.
- The government of the state in which the accident took place is responsible, under Annex. 13 of the Chicago convention for the initiation of an accident investigation.

NATIONAL PROVISIONS

Accident- An occurrence associated with the operation of an aircraft which-

- a) In the case of a manned aircraft, takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked.
- b) In the case of an unmanned aircraft, takes place between the time the aircraft is ready to move with the purpose of flight until such time as it comes to rest at the end of the flight and the primary propulsion system is shut down in which

A person is fatally or seriously injured as a result of:

- Being in the aircraft
- Direct contact with any part of the aircraft
- Direct exposure to jet blast

The aircraft sustains damage or structural strength, performance or flight characteristics of the aircraft to propellers, wing tips, antennas, probes, vanes, tires, brakes, wheels, fairings, panels, landing gear.

Serious injury: An injury which is sustained by a person in an accident and which requires hospitalization for more than 48 hours, commencing within seven days from the date the injury.

Incident: in relation to an aircraft means an occurrence which takes place either on the ground or in flight in which-

- The aircraft suffers damage or a person associated either with the maintenance or operation of aircraft or both suffers injury in circumstances other than those specified in the definition of accident.
- The aircraft makes a forced landing
- The aircraft lands at aerodrome in an un airworthy condition
- The aircraft is compelled to land at the aerodrome of departure without schedule.

Notification of Incidents: Where an aircraft is involved in an incident the Pilot-in-command the owner the operator and the hirer, if any of such aircraft shall notify the incident to the Director-General within 48 hours of the incident. The information shall be sent as soon as possible by the quickest.

Quickest means available and in any case within 24 hours after the occurrence of such incidents.

HUMAN PERFORMANCE AND LIMITATIONS

The partial pressure describes the distribution of certain gases in a mixture and follows Dalton's Law which states that the total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases which compose the mixture.

A pilot, climbing in a non-pressurised aircraft and without using supplemental oxygen will pass the "critical threshold" at approximately 22000ft

Breathing 100% oxygen will lift the pilot's physiological safe altitude to approximately 38000ft.

THE NERVOUS SYSTEM

The brain performs an incredible number of tasks-

- It controls body temperature, blood pressure, heart rate and breathing
- It accepts information about the world around you from your various senses seeing, hearing, smelling, tasting and touching
- It controls decision making and speech through the brain we experience consciousness, vision, hearing, taste, smell, thought and memory

The Nervous System is divided into 3 main parts-

The Central Nervous System

The central nervous system consists of the brain and the spinal cord. Brain, Spinal cord and peripheral nerves make up a complex, integrated information-processing and control system known as Central Nervous System. The spinal cord contains the nerves that carry messages between the brain and the body.

The Peripheral Nervous System – the peripheral nervous system includes all peripheral nerves. It connects the central nervous system to the organs and muscles of the body. The peripheral nervous system controls organs and muscles like skin, eye, blood vessels, heart and stomach.

The organs such as the heart, stomach and intestine are regulated by a part of the nervous system called the Autonomic Nervous System.

The autonomic Nervous System controls eye, heart, breathing, temperature, blood pressure, stomach, intestines and bladder, urinary output, sweating, glands, fight or flight response or reaction to stress.

THE RESPIRATORY SYSTEM

The respiratory process consists mainly of the diffusion of oxygen through the respiratory membranes into the blood, transportation to the cells, diffusion into the cells and elimination of carbon dioxide from the body.

As you breathe air in through your nose or mouth, it goes past the epiglottis and into the trachea. It continues down the trachea through your vocal cords in the larynx until it reaches the bronchi. From the bronchi, air passes into each lung. The air then follows narrower and narrower bronchioles until it reaches the alveoli.

External Respiration

Breathing – normal breathing is a purely automatic process under the unconscious control of the nervous system. The normal rate of respiration in adults 14 to 18 breaths per minute.

THE CIRCULATORY SYSTEM

The circulatory system consists of the heart and the blood vessels and maintains the flow of blood throughout the body.

THE HEART

The arteries carry blood from the heart at high pressure and the veins return blood to the heart at low pressure.

At rest the cardiac output (the quantity of blood the heart pumps in one minute) of an adult is approximately 5 litre/min.

THE CIRCULATORY SYSTEM

When the heart muscle contracts or beats (called systole), it pumps blood out of the heart. The heart contracts in two stages. In the first stage, the right and left atria contract at the same time, pumping blood to the right and left ventricles. Then the ventricles contract together to propel blood out of the heart. Then the heart muscles relax (called diastole) before the next heartbeat. This allows blood to fill up the heart again.

When the body experiences stress or fear, adrenaline is released into the bloodstream causing an immediate increase in the pulse rate.

COMPOSITION AND FUNCTIONS OF THE BLOOD

Blood has two main components – plasma and formed elements.

Formed elements include white blood cells (WBCs), which are part of the immune system, and platelets, which help form clots. White corpuscles produce antibodies to fight bacteria. Platelets are the smallest of the blood cells and assist in the blood clotting process.

Red blood cells (RBCs) are responsible for one of blood's most important tasks carrying oxygen and carbon dioxide.

Inside each RBC are about 280 million haemoglobin molecules.

THE PRINCIPAL FUNCTIONS OF THE BLOOD ARE:

To carry oxygen to and carbon dioxide from the various tissues and organs of the body.

To assist in temperature control of the body.

HYPOXIA

The term cerebral hypoxia technically refers to lack of oxygen supply to the cerebral hemispheres. However, it is more typically used to refer to a lack of oxygen supply to the entire brain.

But the pilot must know that oxygen deprivation or hypoxia can be caused by breathing air at the low pressures at high altitude.

TIME OF USEFUL CONSCIOUSNESS

Altitude	Time of useful consciousness
45,000 feet MSL	9 to 15 seconds
40,000 feet MSL	15 to 20 seconds
35,000 feet MSL	30 to 60 seconds
30,000 feet MSL	1 to 2 minutes
28,000 feet MSL	2 to 3 minutes
25,000 feet MSL	3 to 5 minutes
22,000 feet MSL	5 to 10 minutes
20,000 feet MSL	30 minutes or more

Hypoxia will cause a pilot's intellectual and sensory judgment to become impaired.

Hypoxia is classified into 4 different types

- Hypoxic hypoxia – hypoxic hypoxia is the result of low oxygen levels in the bloodstream. In pilot's this most often occurs with exposure to altitude.
- Anaemic hypoxia – oxygen in blood is carried by haemoglobin which is found in red blood cells. When the red blood cell count decreases or the haemoglobin does not function properly, less oxygen can be carried by the blood. This can occur in conditions such as heavy bleeding or carbon monoxide poisoning. A person suffering from anaemia may notice symptoms such as breathlessness, fatigue or chest pain.
- Ischaemic hypoxia/stagnant hypoxia – the term ischaemia refers to inadequate supply of blood and ischaemic hypoxia occurs when there is inadequate blood flow to body tissues. For e.g. This is often seen in fingers and toes exposed to cold.
- Histotoxic hypoxia – histotoxic hypoxia refers to an inability of the cells of the body to use the oxygen available. It can occur with certain conditions such as cyanide poisoning, chemical poisoning.

SYMPTOMS –

- Impaired judgement
- Headache
- Tingling in hands & feet
- Hyperventilation
- Muscular impairment
- Memory impairment
- Sensory loss
- Tunnel vision

Carbon monoxide poisoning

That is a product of incomplete combustion. Haemoglobin the oxygen-carrying chemical in the blood picks up carbon monoxide over 200 times more readily than it picks up oxygen. Thus, even minute quantities in the cockpit may result in pilot incapacitation.

Carbon monoxide poisoning symptoms –

- initially there is an inability to concentrate
- headache
- dizziness
- nausea
- impaired vision
- lethargy or weakness
- impaired judgment
- personality change
- impaired memory

Hyperventilation

Hyperventilation is another word for over breathing and may be defined as lung ventilation in excess of the body's needs.

Symptoms of Hyperventilation are –

- dizziness
- tingling
- visual disturbances
- hot or cold sensation
- anxiety
- increased heart rate
- spasms
- cramping and spasms of the hands and feet.

Treatment of Hyperventilation-

One of the direct causes of hyperventilation is a reduction in the carbon dioxide level in the blood, the condition may be alleviated by getting the sufferer to breathe into a paper bag. This action will increase the bloods carbon dioxide level causing the brain to reduce the breathing rate.

Smoking – a pilot who is also a smoker may experience the symptoms of oxygen deprivation or hypoxia, at a lower altitude (7,000) than a non-smoker (10,000)

A person who smokes one packet of cigarettes per day will reduce his capacity to carry oxygen by 5-8%.

Alcohol

Alcohol acts primarily as a depressant.

Blood pressure

- blood pressure is the pressure exerted by blood on the walls of the main arteries.
- The permanent pressure against the arterial wall is called diastolic pressure. The increased pressure occurring with each beat of the heart is called the systolic pressure.
- 120/80 is a normal blood pressure for a healthy young adult.

Causes of blood pressure (hypertension)

- Stress
- Smoking
- Poor diet
- Obesity
- Lack of exercise
- Age
- Narrowing of the arteries

Decompression sickness

Henry's Law explains the occurrence of decompression sickness. Symptoms of decompression sickness are bends, chokes, skin manifestations, neurological symptoms and circulatory shock. Do not fly for at least 12 hours after experiencing rapid decompression even though you may be feeling fit.

Flying following diving

As a general rule, individuals should not fly within 24 hours following diving and certainly not the same day.

FITS AND FAINTS

Epilepsy

A fit or seizure is usually referred to as epilepsy. A fit or a seizure is not a specific disease but a set of signs or symptoms in response to a disturbance of the electrical activity in the brain.

PHYSIOLOGY

Physiology is a technical term meaning the science of the organic functions of humans, animals and plants.

The retina – the retina has rods on its peripheral zone and cones on its central zone. The retina contains the receptors for vision: about 100 million rods and 6 million cones.

The rods – the rods can only detect black and white but are much more sensitive at lower light levels. Rods are responsible for our peripheral vision.

LIMITATIONS OF THE VISUAL SYSTEM

Light vision – Adaptation

Adaptation is the adjustment of the eyes to high or low levels of illumination. The time required for complete adaptation for high levels of illumination is 10 sec and for full dark adaptation 30 min. Dark adaptation takes time – about 7 minutes for the cones and 30 minutes for the rods.

The Eustachian Tube-

The eustachian tube allows pressure in the middle ear to equalize across the ear drum with outside or ambient pressure when climbing or descending. No one should fly if their Eustachian tube is blocked and they cannot clear their ears.

Otis Barotraumas – stretching of the ear drum caused by the expansion and contraction of gases trapped in the inner ear by a blocked eustachian tube.

Audible Range of the Human Ear – the range of pitch or frequency of sounds that a fit young person can hear lies between 20 and 20,000 Hertz, or cycles per second.

Noise Induced Hearing Loss – any prolonged exposure to noise in excess of 90 dB can end up in noise induced hearing loss. This can damage the very sensitive membrane in the cochlea.

Orientation- human beings maintain spatial orientation using a combination of 3 factors-

- The sense of vision
- The Vestibular Apparatus
- The Somatosensory system (seat of the pants feeling)/G-Force

The most reliable sense is the sense of vision.

Illusions when taxiing – relative movement

When taxiing into a headwind the blowing snow will give the illusion that the aircraft is taxiing faster than it actually is.

Illusion on take-off – somatogravic illusion

An acceleration gives the pilot an impression of the nose of the aircraft pitching up.

Width of the runways – A pilot used to a standard width runway may, when approaching an unfamiliar narrow runway, judge he is too high and therefore round out too low on approach.

- Shallow Approaches: upslope runway or terrain, narrower than usual runway, feature less terrain, rain on the wind screen and haze give an illusion of being high resulting in shallower approaches.
- Steep Approaches: down sloping runway or terrain, wider than usual runway and bright runway/app.lights give illusion of being low, resulting in high approaches.

Disorientation/ Vertigo

If you suspect disorientation, concentrate on and believe in aircraft instruments in IMC. If in VMC look out at the horizon.

Air/motion sickness- this mismatch between vestibular and visual sensory input is the primary cause of spatial disorientation and indeed of motion sickness.

The Leans – an abrupt correction of a banked attitude, which has been entered too slowly to stimulate the motion sensing system in the inner ear, can create the illusion of banking in the opposite direction.

Somatogravic illusion – a rapid acceleration during take-off can create the illusion of being in a nose up attitude. The disoriented pilot will push the aircraft into a nose low, or dive attitude.

Disorientation summary

- Abrupt head movement during a prolonged constant rate turn in IMC or simulated instrument conditions can cause pilot disorientation
- A sloping cloud formation an obscured horizon and a dark scene spread with ground lights and stars can create an illusion known as false horizons.
- An abrupt change from climb to straight and level flight can create the illusion of tumbling backwards.
- A rapid acceleration during take-off can create the illusion of being in a nose up attitude
- Symptoms of hypoxia may be difficult to recognize before the pilot's reactions are affected.

Nutrition and food hygiene – never fly on an empty stomach. A balanced diet is the foundation of good health.

CAFFEINE

Caffeine is probably the most widely used drug in the world. It can easily lead to addiction. Caffeine is present in coffee, tea, cocoa, chocolate and fizzy drinks such as cola. The recommended maximum caffeine intake per day is approximately 250-300 mg corresponding to 2-3 cups of coffee.

Hypothermia – normal body temperature averages 98.6 degrees. With hypothermia, core temperature drops below 95 degrees. In severe hypothermia core body temperature can drop to 82 degrees or lower. Hypothermia affects physical and mental abilities.

Acceleration and G Forces – on the ground the body is subject to normal gravitational acceleration “1G” this is 32 ft/sec or 9.81 m/sec.

At 60% of bank a pilot is subject to an acceleration of 2G acting vertically through his seat. His weight will also increase by a factor of 2. This is called the load factor. In a 70-degree level turn, the load factor increases to 3.

During increased G manoeuvres the pilot will feel the following effects –

- Increase in body weight
- Mobility is impaired
- Internal organs displaced
- Onset of tunnel vision

1.41G is the acceleration experienced in a level turn at 45 degrees angle of bank.

Negative G – most pilots can learn to tolerate moderate increases in positive G, but many find even the smallest exposure to negative G to be unpleasant. Negative G manoeuvres increase the flow of blood to the head. Blood pressure there increases.

Crew Resource management

CRM- is the effective use of all available resources for flight crew personnel to assure a safe and efficient operation, reducing error, avoiding stress and increasing efficiency.

Cognitive and interpersonal skills needed to manage the flight within an organised aviation system. In this context cognitive skills are defined as the mental processes used for gaining and maintaining situational awareness. For solving problems and for taking decisions. Interpersonal skills are regarded as communications and a range of behavioural activities associated with teamwork.

Basic information processing – a functional model

We need to build a functional model for the various stages of our reasoning. these stages are –

- Detection
- Perception
- Decision making
- Action
- Feedback

The brain, central decision maker and response selection-

Once information has been perceived a decision must be made as to the response. Information is continuously entered into and withdrawn from both the long and short-term memories to assist the decision process. To enable us to carry out multi-tasks we must learn skills through Motor Programme.

Motor Programmes (skills) –

Motor Programmes or “Skills” are behavioural sub-routines which are learnt by practice and/or repetition and are held within the long Term Memory and can be carried out without conscious thought.

Developing Motor Programme

There are 3 distinct phases to developing a Motor Programme or skill:

- a) The cognitive phase
- b) The associative phase
- c) Th automatic phase

CONCEPTS OF SENSATION

Stimuli

Types of Memory – there are 3 types of memory

- a) Sensory memory
- b) Short-term memory
- c) Long-term memory

AROUSAL

Arousal is a major aspect of many learning theories and is closely related to other concepts such as anxiety, attention, agitation, stress and motivation.

Low Arousal – at times such as in the cruise, out attention can wander with the result that information is either missed or misinterpreted.

SYMPTOMS OF OVERLOAD

The most common symptoms are:

- a) A sharp degradation of performance
- b) Funnelling of attention or focus
- c) Regression
- d) Mental blocking
- e) Mood swings
- f) Restlessness and trembling
- g) Panic

Human reliability- the rate of human error during simple and repetitive tasks might be expected to be 1 in 100, but after practice, a rate of 1 in 1000 could be achieved.

Short term memory (working memory) 10 to 20 seconds. That information will be lost in 10 to 20 seconds unless it is actively rehearsed and deliberately placed in our long-term memory.

Limitations of Short Term Memory – the maximum number of unrelated items which can be maintained in the short-term memory is about 7±2. Once this limit is exceeded one or more of the items are likely to be lost or transposed.

LONG TERM MEMORY

Introduction – it is believed that information is stored in the Long-term Memory for an unlimited time period, although frequently there may be retrieval problems.

- Semantic memory – semantic memory stores general knowledge of the world.
- Episodic memory – episodic memory is a memory of events or episodes in our life.
- Procedural memory – skills are included within the make-up of the Long-Term Memory.

Motivation – motivation is the combination of a person's desire and energy directed at achieving a goal. It is the cause of action. Motivation can be intrinsic, such as satisfaction and feelings of achievement or extrinsic, such as rewards, punishment, and goal obtainment. The learning process is vastly improved with motivation, and high performance is rarely achieved without it.

Experience - we all have the ability to learn from our experiences and mistakes and from those of others.

Judgment & decision making – making good decision is one of the important aspects of piloting and aircraft. Good decision can help lead to safe, successful flights. Whereas bad decisions and even indecision have led to many aircraft accidents. When flying with another pilot, it is possible to increase the chance of a good decision, being made by discussing the potential problems beforehand. If as an inexperienced pilot you are flying with someone of more experience and you see him doing something you consider to be dangerous, you should immediately question their course of action.

A good pilot is one who always realizes there is more to learn.

Risk Assessment- risk assessment is based on the probability of the risk occurring and the impact of that risk if it did occur, you will have to weigh up risk and the possible consequences of the risk in order to determine your course of action.

Personality, Attitude and Behaviour – someone who is too introvert or anxious may not make a good pilot. As captain of an aircraft you will need to show good leadership skills.

While acting as pilot-in-command you should ensure that you stay in command, you should be properly organized and prepared.

Situational Awareness – a good leader displays good situational awareness and accurately assess his own performance. The following factors can interfere-

- 1) Stress
- 2) Boredom
- 3) Fatigue
- 4) Emotional disturbance
- 5) Poor communication
- 6) Interruptions

Mechanics of decision-making – pilots operate in a dynamic and constantly changing environment. A good decision reached a minute ago will not necessarily be the same good decision in 2 minutes time.

- Recognize and identify the problem
- Consider the nature of the problem that you are trying to resolve
- Analyse or research the problem
- Develop a list of possible solutions
- Select the best alternative
- Execute the best choice
- Follow up and communication
- Feedback

Communication – concise and unambiguous communication is essential to the safe conduct of air traffic.

Barriers to communication and Team Work

Sometimes certain attitudes on the part of both leaders and team members build barriers to communication and can be especially dangerous when flying. Some of the barriers to communication and team work are –

- Aggressiveness
- Arrogance
- Anti- authoritarian behaviour
- Impulsiveness
- Feelings of invulnerability
- Resignation

Stress Tolerance and Arousal

- The response of a person to the event or circumstances to which he is exposed is known as arousal.
- Personality, character, training and experience will influence levels of arousal in a given individual.

Stress Factors

Stress causes mental blocks, confusion, channelized attention, resignation, frustration, rage, deterioration in motor coordination, high pitch voice and fast speaking.

Thorough flight planning is the key to reducing cockpit workload

Make sure that cockpit housekeeping is of a high order and that all documents, charts and associated equipment are appropriately stored and accessible, in accordance with the principles of flight safety.

High quality headsets will reduce cockpit noise levels.

To avoid stress, know your limits and fly well inside them.

Life stress

Pilots suffering from life stresses should be aware that this can affect their concentration and performance when at the controls of an aircraft. The descending order in which the factors affect a person are – death of spouse/child, divorce, marital separation, death of a close family member, injury/illness, marriage, loss of job, retirement, pregnancy, sexual problems, birth, change of financial status, siblings leaving home, change of eating habits, change of residence, loan/mortgage, vacations and minor violations of law.

Indications of stress- easily observable indications of stress are perspiration, flushed skin, dilated pupils and fast breathing.

THE CIRCADIAN CIRCLE

The Circadian Circle represents our level of alertness throughout the day. Circadian clock functions as the main control centre for the autonomic nervous system by regulating sleep cycles, body temperature appetite etc. Body temperature, heart rate and blood pressure effect our level of alertness during the course of a day.

Pilots flying across different time zones through the night, are not able to fully apply the circadian cycle without understanding or adjusting it. By understanding how our alertness is affected, we can then make adjustments to our diet and sleep to ensure a safe level of alertness in the cockpit.

SLEEP - sleep is basically divided into two components. Rapid Eye Movement (REM) sleep and Non-Rapid Eye Movement (NREM) sleep. The purpose of NREM sleep is body restoration while that of the REM sleep is brain restoration by strengthening, refreshing and organizing memory.

NREM SLEEP – STAGE 1 – the transition phase between wakefulness and sleep takes 10 minutes each time. The brain activity, eye movement and muscle activity become slower. A person is easily awoken. Waking up in this stage of sleep will cause a person to feel that he/she has not slept.

NREM SLEEP – STAGE 2 – light sleep, first stage of true sleep lasts from 10-25 minutes each time. Occupies 50% of the sleep patterns, the brain activity, eye movement become even slower and the cardiac activity decreases.

NREM SLEEP – STAGE 3 – the beginning of deep sleep, slow-wave delta sleep. The brain activity and eye movement are approaching zero. If a person is awoken in this stage of sleep, he/she may feel groggy and disoriented for a few minutes.

NREM SLEEP – STAGE 4 – deep sleep, slow-wave delta sleep. No eye movement or muscle activity. If a person is awoken in this stage of sleep, he/she may feel groggy or disoriented for a few minutes.

REM SLEEP (Paradoxical sleep) – usually occurs 70-90 minutes into a sleep. Happens in cycles lasting for approximately 10 minutes initially to 1 hour. The eye moves rapidly from side to side and the brain wave form is similar to that when a person is awake. The dreams occur. As we grow older, the time spent in REM sleep declines from 50% of our sleep for infants to 20% of our sleep for adults.

- The effects of sleep deprivation on performance increase with altitude and with higher workload.
- During paradoxical sleep rapid eye movements can be observed.
- Deep sleep allows for physical recovery and the reconstitution of neuron energy reserves.
- Lack of sleep increases fatigue, concentration and attention. It has risks of sensory illusions and mood disorders. The disruption of sleeping patterns can lead to symptoms of drowsiness, irritability and lack of concentration will make an individual more prone to make errors.

THE SHEL MODEL IN AVIATION

The SHEL model provides a conceptual framework to help understand Human Factors. Human factors elements can be divided into 4 basic conceptual categories –

S- Software- procedures, manuals, checklist layouts, documents, zymology, computer programmes, maps and charts.

H- Hardware – design of flight decks, physical structure of the aircraft, presentation of instruments, positioning and operating sense of controls, machinery, equipment etc.

E- Environment – both internal and external to the workplace

L- live ware – man, the most critical, valuable and flexible component of the system.

L- live ware – the second L represents other humans in contact with the pilot.

Interaction between human beings and other elements of the SHEL model are at the heart of Human Factors which involves the interface between –

- Humans and machines “live ware-Hardware”
- Humans and materials “live ware-Software”
- Humans and their colleagues “Live ware – live ware”
- Humans and the operating environment “live ware – environment”

The live ware (human element) is the hub of the SHEL model. The remaining components must be adapted and matched in this central component.

DEFINITIONS

Accelerate Stop Distance Available (ASDA). The length of the take-off run available plus the length of the stopway, if provided.

Advisory Route. A designate route along which air traffic advisory service is available.

Advisory Service. Advice and information provided by a facility to assist pilots in the safe conduct of flight and aircraft movement.

Aircraft Approach Category (AAC). A grouping of aircraft based on a Ground speed of 1.3 times the stall speed in the landing configuration i.e; Vat at maximum gross landing weight. If it is necessary to manoeuvre at speeds in excess of the upper limit of a speed for a category, the minimum for next higher category should be used.

Category A- Speed less than 91 Knots

Category B- Speed 91 Knots or more but less than 121 Knots

Category C- Speed 121 Knots or more but less than 141 Knots

Category D- Speed 141 Knots or more but less than 166 Knots

Category E- Speed 166 Knots or more

Air Defence Identification Zone (ADIZ). The area of airspace over and or water, extending upward from the surface, within which the ready identification, the location and the control of aircraft are required in the interest of national security.

Airport Reference Point (ARP). A point on the airport designate as the official airport location.

Air Traffic Control Clearance. An authorization by air traffic control, for the purpose of preventing collision between known aircraft, for an aircraft to proceed under specified traffic conditions within controlled airspace.

Airway. A control area or portion thereof established in the form of a corridor equipped with radio navigation aids.

Alternate Aerodrome. An aerodrome to which an aircraft may proceed which it becomes either impossible or inadvisable to proceed to or to land at the aerodrome of intended landing. Alternate aerodromes include Take-off alternate, En-route alternate and Destination alternate. An alternate aerodrome may be the aerodrome of departure.

Area Navigation (RNAV). A Method of navigation which permits aircraft operation on any desired flight path within the coverage of the station-referenced navigation aids or within the limits of the capability of self-contained aids or a combination of these.

ATS Route. A specified route designated for channeling the flow of traffic is necessary for the provision of air traffic services.

Automatic Terminal Information Services (ATIS). The provision of current routine information to arriving and arriving aircraft by means of continuous and repetitive broadcast throughout the day or specified portion of the day.

Ceiling. The height above the ground or water of the base of the lowest layer of cloud below 6,000 meters (20,000) covering more than half the sky (4 Octa).

Circling Approach. An approach during which an aircraft manoeuvres, outside of the 30° arc of the centreline of the active runway, in order to achieve a suitable position from which it can continue with a visual approach.

Clearway. A defined rectangular area on the ground or water under the control of the Appropriate Authority, selected or prepared as a suitable area over which an aircraft may make a portion of its initial climb to a specified height.

Control Area. A controlled airspace extending upwards from a specified limit above the earth.

Control Airspace. A controlled airspace of defined dimensions within which air traffic control services is provided flights in accordance with airspace classification.

Control Zone. A controlled airspace extending upwards from the surface of the earth to a specified upper limit.

Current Flight Plan. The flight plan including changes, if any, brought about by subsequent clearance.

Decision Altitude/Height (DA/DH). The specified altitude, on height in the precision approach at which a missed approach must be initiated if the required visual references to continue the approach has not been established.

DA is referenced to mean sea level and DH is referenced to threshold elevation

The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path

Distress Phase. A situation wherein there is a reasonable certainty that an aircraft and its occupants are threatened by grave and imminent danger or require immediate assistance.

Displaced Threshold. A threshold that is located at a point on the runway other than designated beginning of the runway.

Final Approach. That part of an instrument approach procedure which commences at the specified final approach fix or points or where such a fix or not specified

Flight information centre. A unit established to provide flight information service and alerting service.

Flight information region an airspace of defined dimensions within which flight information useful for the safe and efficient conduct of flights.

Flight Plan. Specified information provided to Air Traffic Service Units, relative to the intended flight or portion of a flight of an aircraft.

Grid Minimum Off Route altitude (GRID MORA). An altitude derived by Jeppesen or provided by State authorities. The grid MORA altitude provides terrain and manmade structure clearance within the section outlined by latitude and longitude lines. MORA does not provide for NAV AID signal coverage or communication coverage.

Holding Point. A specified location, identified by visual or other means, in the vicinity of which the position of an aircraft in flight is maintained in accordance with air traffic control instruction.

Initial Approach. That part of an instrument approach procedure in which the aircraft has departed an initial approach fix or point and is manoeuvring to enter the intermediate or final approach. It ends at the intermediate fix or point or, where no intermediate segment is established, at final approach fix point.

Instrument Approach Procedure. A series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix or where applicable, from the beginning of a defined arrival to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or enroute obstacle clearance apply.

Landing Distance Available (LDA). the length of runway which is declared available and suitable for the ground run of an aircraft landing. The landing distance available commences at the threshold and in most cases corresponds to the physical length of the runway pavement.

Maximum Authorised Altitude (MAA). A published altitude representing the maximum usable altitude of flight level for an airspace structure of route segment.

Minimum Obstruction Clearance Altitude (MOCA). The lowest published altitude in effect between radio fixes on VOR airways, off airway routes or route segments which meets obstacle clearance requirements for the entire route segments.

Minimum Off Route Altitude (MORA). This is an altitude derived by Jeppesen. The MORA provides known obstruction clearance 10 NM either side of the route centreline including a 10 NM radius beyond the radio fix reporting or mileage break defining the route segment.

Minimum Enroute Altitude (MEA). The lowest published altitude between radio fixes that meets obstacle clearances requirements between these fixes and in many countries assures acceptance navigational single coverage.

Minimum Descent Altitude or Height (MDA/DH). A specified altitude or height in a non-precision approach or circling approach below which descent may not be made without visual references.

Minimum Safe Altitude (MSA). The lowest altitude which may be used under emergency conditions which will provide a minimum clearance of 800 meters (1000 ft) above all obstacles located in an area centred with a sector of a circle of 25 nautical miles radius centred on a radio aid to navigation. This altitude is for emergency use only and does not necessarily guarantee NAV AID reception. When the MSA is divided into sector, with each sector a different altitude, the altitudes in these sectors are referred to as "minimum sector altitude".

Missed Approach Point (MAP). The point in an instrument approach at or before which the prescribed missed approach procedures must be initiated in order to ensure that minimum obstacle clearance is not infringed.

Non-Precision Approach. An instrument approach using a procedure which does not employ electronic glide-slope information e.g. VOR, TACAN, ASR etc.

Notice to Airmen (NOTAM). A notice containing information concerning the establishment condition or change in any aeronautical facility service, procedures of hazard, the timely knowledge of which is essential to personnel concerned with flight operations.

Obstacle Clearance Altitude/ Height (OCA/OCH). The lowest altitude or alternatively the lowest height above the elevation of the relevant runway threshold or above the aerodrome elevation as applicable (OCH) used in establishing compliance with appropriate obstacle clearance criteria (OCC).

Precision Approach. An instrument approach using a facility which provides both azimuth and electronic glide-slope information.

Precision Approach radar. Primary radar equipment used to determine the position of an aircraft during final approach, in terms of lateral and vertical deviations relative to a nominal approach path and in range relative to touchdown.

Procedure Turn. A manoeuvre in which a turn is made away from a designated track followed by a turn in the opposite direction to permit the aircraft to intercept and proceed along with reciprocal of the designated track.

- Procedure turns are designated "left" or "right" according to the direction of the initial turn.
- Procedure turns may be designated as being made either in level flight or while descending, according to the circumstances of each individual procedure.

Restricted Area. An airspace of defined dimensions, above the land areas or territorial waters of a state, within which the flight of aircraft is restricted in accordance with specified conditions.

SELCAL. A selective call system which permits the selective calling of individual aircraft over radio-telephone channels linking a ground station with the aircraft.

Side Step Manoeuvre. A visual manoeuvre accomplished by a pilot at the completion of an instrument approach to permit a straight in landing on parallel runway, not more than 1200 feet to either side of the runway to which instrument approach was conducted.

Special VFR Flight. A controlled VFR flight authorised by air traffic control to operate within a control zone under meteorological conditions below the visual meteorological conditions.

Standard Instrument Arrival (STAR). A designated instrument flight rule (IFR) arrival route linking significant point., normally on an ATS route,, with a point from which a published instrument approach procedure can be commenced.

Standard Instrument Departure (SID). A designated instrument flight rule (IFR) departure route linking the aerodrome or a specified runway of the aerodrome with a specified point, normally as a designated ATS route, at which the enroute phase of flight commences.

Stop way. A define rectangular area on the ground at the end of the take-off run available prepared as a suitable area in which an aircraft can be stopped in the case of an abandoned take-off. A stop way should have the same width as the runway with which it is associated.

Take-off Distance Available (TODA). The length of the take-off run available plus the length of clearway available (if clearway is provided), OD is not to exceed $1.5 \times \text{TORA}$.

Take-off Run Available (TORA). The length of runway declared available and suitable for the ground run of an aircraft taking off. This is most causes, corresponds to the physical length of runway pavement.

Terminal Control Area. A control area normally situated at the confluence of ATS routed in the vicinity of one or more major aerodromes.

Threshold. The beginning of that portion of the runway usable for landing.

Transition Altitude. The altitude in the vicinity of an aerodrome at or below which the vertical position of an aircraft is controlled by reference to altitude above mean sea level or height above the aerodrome depending on whether QNH or QFE is set on the altimeter.

Transition Layer. The airspace between the transition altitude and the transition level. The depth of this layer will normally be insignificant and will in any is never exceed 500 feet.

Transition Level. The lowest flight level above the transition altitude. It will vary in accordance with the relationship between the QNH and the standard pressure datum.

Visual Approach. An approach by and IFR flight when either part on all of an instrument approach procedure is not completed and the approach is executed in visual reference to the terrain.

Wake Turbulence. The term “wake turbulence” used to describe the effect of the rotating air masses generated behind the wing tips of large jet aircraft, in preference to the term “wake vortex” which describes the nature of the air masses.

Wake Turbulence Categorisation of Aircraft. Wake turbulence separation minima shall be based on a grouping of aircraft types into three categories according to maximum certificate take-off mass as follows:

Heavy (H). all aircraft types of 136,000 kg or more

Medium (M). aircraft types less than 136000 kg but more than 7000 kg

Large (L). aircraft types of 7000 kg or less

Way Point. The specified geographical location used to define an area navigation route of the flight path of an aircraft employing area navigation.